

Quoting Under a Misspecified Fill Law: Order-Book Queues, Adverse Selection, and Inventory-Control Certificates

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Abstract

A market maker chooses prices and messages using an uncertain law of fills, queue movements and post-fill price changes. This paper connects errors in that law to the value of a deployed quoting policy. A marked-wealth identity separates spread capture, carried-inventory drift and simultaneous fill-price jumps. A finite event model retains tagged queue priority, partial fills, inventory reservations and orders that remain executable while cancellation is pending. A finite-stage latency approximation has an explicit leading value error and remainder.

The control certificates account for primitive model errors, continuation values and numerical residuals. Smooth quote adjustments admit quadratic bounds; discrete messages require action gaps and state-dependent occupation bounds. A baseline-relative extension turns an enclosure of the current policy's value into a selective improvement rule. It changes a message only when its advantage is certified nonnegative throughout the uncertainty set, and supplies a computable lower bound on the resulting discounted gain.

A marginal-coupling calculation gives a sharp participation interval when fill and price-jump dependence is unknown. A finite-message implementation layer encloses rounding and score errors, retains cancellation tails and states a joint-event calibration contract. The monograph's finite-sample interface transfers supported calibration uncertainty and compiled-policy error to the same objective.

A continuing 74-state queue-and-inventory experiment verifies both loss and improvement certificates with rational outward enclosures and independent policy evaluation. Additional lower bounds show the cost of omitting tagged rank and the difficulty of calibration near a replacement boundary. These are conditional mathematical guarantees, supported by synthetic calculations; they do not establish empirical fill calibration or trading profitability.

Keywords: market making; adverse selection; marked point processes; queue priority; inventory control; model misspecification.

Contents

1	Introduction	4
2	Wealth accounting with simultaneous fills and price jumps	5
3	A finite-state model of inventory control	6
3.1	States, actions, and rewards	6
3.2	Existence, verification, and the policy loss identity	7
3.3	Book events, tagged orders, and admissible states	8
3.4	Queue priority and competing execution times	9
3.5	Batch consumption, partial fills, and marked volume	11
3.6	Cancellation and replacement at decision opportunities	13
3.7	Live orders during cancellation latency	13
3.8	From event-law errors to Hamiltonian error budgets	15
3.9	The cost of truncating the book	16
4	Offsets, inventory shadow values, and selection effects	17
5	An exact gross-profit shutdown threshold	18
6	Certificates for a misspecified quoting model	19
6.1	A constrained local certificate	19
6.2	From a quote error to total policy regret	20
6.3	Primitive errors and the continuation-value contribution	20
6.4	Withdrawal, ticks, and branch changes	21
6.5	Stationary discounting and the effective horizon	22
7	A certificate from computed values and supplied error bounds	23
7.1	A discrete-message certificate from value intervals	25
8	Improving a baseline quoting policy under uncertainty	26
8.1	An advantage measured against one common law	26
8.2	The message-clock specialization	27
8.3	A queue-aware baseline experiment	28
9	Hybrid actions and the cost of a small decision margin	29
10	Why a fill certificate requires action coverage	30
10.1	A positive certificate from two logging quotes	31
10.2	Information and sampling lower bounds inside a queue	33
11	Signal-value ceilings for bounded and unbounded rewards	34
11.1	A bounded reward	34
11.2	A centered exponential-moment condition	35
12	Synthetic verification	36
12.1	A marked inventory model	36
12.2	An exact enclosure for a computed HJB approximation	37
12.3	Discounting, two-quote learning, and unbounded rewards	38

12.4	Queue mechanics and event-law checks	38
12.5	A continuing queue-and-inventory certificate	39
12.6	Partial fills during a pending cancel	40
13	Beyond a distance-only fill law	42
13.1	The six requirements for a continuing quote	43
13.2	One event, one marked-wealth charge	43
13.3	Sharp ambiguity from missing fill–jump dependence	43
13.4	What would make the added realism measurable?	44
14	From a quoting certificate to an executable message policy	45
14.1	A joint law for messages, fills and price marks	45
14.2	Ticks and implementation certificates without smoothness	45
14.3	The economic value of latency tails	46
14.4	Calibration with competing events and policy selection	47
14.5	Reference implementation and falsification cases	47
14.6	Persistence, competition and the scope of the dealer law	48
15	Calibration, implementation error, and the evidence boundary	48
16	Scope and consequences	49

1 Introduction

Market making combines a quoting decision with an inventory decision. A quote determines which counterparties trade, the inventory they leave, and the information their trades reveal. These effects make the fill law more than a demand curve: it is also part of the controlled transition law and the distribution of post-fill price changes.

Inventory models descend from [Ho and Stoll \(1981\)](#), and adverse-selection models from [Glosten and Milgrom \(1985\)](#). The quoting frameworks of [Avellaneda and Stoikov \(2008\)](#), [Guéant et al. \(2013\)](#), and [Cartea et al. \(2015\)](#) explain how execution probabilities interact with inventory costs. Model uncertainty in arrival rates, fill probabilities, and price dynamics is explicitly studied by [Cartea et al. \(2017\)](#). Thus neither uncertain fills nor inventory-sensitive quotes are new modeling ingredients. More directly, [Cao et al. \(2026\)](#) obtain a second-order performance gap and a logarithmic-squared learning-regret bound for an ergodic Avellaneda–Stoikov model. Quadratic sensitivity of a market-making policy is therefore also an established phenomenon.

The question developed here is a quantitative transfer question: given a policy optimized under one fill-and-price law, what bounds its performance under another? A one-step quote certificate is insufficient for this purpose. The estimated law changes the inventory shadow price as well as the immediate reward, and the policy visits states according to the true controlled dynamics.

Section 3.3 makes the book interpretation explicit. It distinguishes a displayed queue from the position of a tagged order, retains live orders during pending cancellation, and models replacement as a state reset. Proposition 3.12 maps errors in event rates and conditional price marks to the primitive budgets used by the control certificates. Proposition 3.13 adds a separate allowance for exiting a finite retained cell. Queue absorption methods themselves are established ([Cont et al., 2010](#)); their role here is to identify which execution law, state variables, and approximation errors the misspecification comparison must retain. Section 3.5 extends this construction to batch consumption and partial fills, preserving the association between executed volume and its price mark. Theorems 3.10 and 3.11 then retain fills during pending cancellation and quantify the value error of a finite-stage delay model. The latter gives a mean-matched expansion with a sharp k^{-1} leading order, rather than identifying small timing variance with negligible economic loss.

Theorem 6.2 addresses this distinction. It combines a finite-state performance-difference identity with a constrained strong-concavity estimate. Corollary 6.4 then bounds the gradient discrepancy using errors in primitive rewards and transition rates, including the resulting error in continuation values. Theorem 7.1 extends this comparison to a numerically computed value function and an inexact optimizer. It uses supplied primitive uncertainty bounds, an HJB residual, a terminal discrepancy, and a verified curvature enclosure; its right side does not require the true value function. Theorem 9.1 adds a separate occupation-margin term for hybrid actions. Proposition 10.1 shows why arbitrarily much data at one quote cannot by itself justify vanishing uncertainty away from that quote. Proposition 10.2 provides the positive counterpart under a two-quote Poisson experiment. Theorem 6.6 gives the stationary discounted extension, and Corollary 7.2 retains numerical errors in that setting. For discrete quote messages, Theorem 7.3 encloses model-specific values and then evaluates a state-dependent upper bound on the deployed policy’s action gaps. This avoids discarding the queue states in which an error can occur. Section 12.5 implements that comparison with rational arithmetic in a continuing 74-state inventory model. Proposition 10.4 and Theorem 10.5 give complementary lower bounds: one for omitted tagged rank and one for finite calibration near a replacement boundary.

The finite-horizon and discounted certificates concern marked rewards and general controlled rates. The two-quote experiment measures post-calibration revenue loss; it does not supply an online learning algorithm or improve the ergodic learning rate of [Cao et al. \(2026\)](#).

Three distinctions organize the analysis. Carried-inventory drift and fill–price co-jumps are separate accounting terms. A local quote Hamiltonian and a total policy objective are separate optimization objects. Finally, a smooth quote adjustment and a discrete decision to withdraw a quote have different stability requirements.

The execution monograph (Noguer Alonso, 2026) studies a parent mandate, its conserved ledger, and error certificates for feasible execution policies. This paper concerns continuing liquidity provision: there is no prescribed quantity to complete, and inventory is a state whose cost enters the objective. The common object is a feasible policy evaluated under a specified information class. Here the accounting needed for that comparison includes simultaneous fills and price jumps, and the stability calculation separates smooth offsets from discrete participation decisions.

Section 13 organizes six practical modeling requirements, specifies a joint event-accounting boundary, and derives a sharp participation interval when fill and jump marginals are known but their dependence is not.

Implementation extension. Section 14 completes the discrete implementation interface. Section 15 applies the monograph’s finite-sample coverage, policy-transfer bounds and the supported-counterfactual evidence boundary. All new experiments are synthetic, and the new guarantees are conditional on the assumptions stated in those sections.

2 Wealth accounting with simultaneous fills and price jumps

Fix a finite horizon T and a filtered probability space satisfying the usual conditions. The reference price is a special semimartingale

$$S_t = S_0 + M_t + A_t, \quad M_0 = A_0 = 0,$$

where M is a local martingale and A is predictable with finite variation. Unit ask and bid fills are counted by N^a, N^b , with predictable intensities Λ_t^a, Λ_t^b . Predictable offsets $\delta_t^a, \delta_t^b \geq 0$ generate cash and inventory

$$dK_t = (S_{t-} + \delta_t^a) dN_t^a - (S_{t-} - \delta_t^b) dN_t^b, \quad q_t = q_0 + N_t^b - N_t^a.$$

Wealth is $W_t = K_t + q_t S_t$. Quotes are priced relative to the pre-jump reference S_{t-} .

Assumption 2.1 (Integrable admissible policies). Offsets and intensities are bounded, q is bounded, and

$$\mathbb{E}[M]_T < \infty, \quad \mathbb{E} \text{Var}_{[0,T]}(A) < \infty.$$

Counting processes are nonexplosive, have unit jumps, and have no common ask–bid jumps. An inactive side has intensity zero; monotonicity of intensity in its offset is required only on an active side when used later. Inventory constraints are enforced through admissible order states: a side whose execution would violate the bound must already be inactive.

The last convention permits a bounded-inventory model. It cannot be implemented by suppressing the execution of an order that remains live. The reservation invariant in Section 3.3 gives an explicit admissible-state construction, including pending messages.

Theorem 2.2 (Three-term wealth decomposition). *Under Assumption 2.1,*

$$\begin{aligned}\mathbb{E}[W_T - W_0] &= \mathbb{E} \int_0^T (\delta_t^a \Lambda_t^a + \delta_t^b \Lambda_t^b) dt + \mathbb{E} \int_0^T q_{t-} dA_t \\ &\quad + \mathbb{E} \sum_{0 < t \leq T} \Delta q_t \Delta S_t.\end{aligned}\tag{1}$$

All terms in this equality are finite.

Proof. Integration by parts gives

$$dW_t = \delta_t^a dN_t^a + \delta_t^b dN_t^b + q_{t-} dS_t + d[q, S]_t.$$

The process q has finite variation and is purely discontinuous, hence $[q, S]_T = \sum \Delta q_t \Delta S_t$. Bounded predictable integrands and $\mathbb{E}[M]_T < \infty$ make $\int q_{t-} dM$ a square-integrable martingale. Its expectation is zero. The counting-process integrals have the stated compensators.

To check absolute integrability of the bracket, put $N = N^a + N^b$. Then $|\Delta q| \leq 1$ and

$$\sum_{t \leq T} |\Delta q_t \Delta M_t| \leq \sqrt{N_T} \sqrt{[M]_T}.$$

Its expectation is finite by Cauchy–Schwarz, since bounded intensities give $\mathbb{E}N_T < \infty$. The contribution involving A is bounded by $\text{Var}_{[0,T]}(A)$, and the drift integral by $\sup |q| \text{Var}_{[0,T]}(A)$. Taking expectations proves (1). $\square$

Corollary 2.3 (A martingale reference). *If S is a martingale, the drift pairing in (1) is zero. Expected wealth then equals offset capture plus the expected co-jump term. It equals offset capture alone if that expected co-jump term is zero, in particular if fills and price jumps never occur together.*

Example 2.4 (A martingale with a fill loss). Let N be a rate- λ Poisson process and $\tau = \inf\{t : N_t = 1\}$. Consider one ask fill at τ , then withdraw the quote. Set

$$q_t = -N_{t \wedge \tau}, \quad S_t = j(N_{t \wedge \tau} - \lambda(t \wedge \tau)), \quad j > 0.$$

The stopped compensated Poisson process is a martingale and inventory lies in $\{-1, 0\}$. At the fill, $\Delta q = -1, \Delta S = j$, so

$$\mathbb{E}[W_T - W_0] = (\delta - j)(1 - e^{-\lambda T}).$$

The reference is a martingale, but the maker loses j per filled unit relative to gross offset capture.

For a simultaneous ask fill and price jump J , the co-jump contribution is $-J$; for a bid fill it is $+J$. A price move occurring strictly after the fill instead affects the inventory carried over that later interval. Changing the marked reference changes the offset definition as well; it does not remove a term while leaving the cash equation fixed. Inventory cannot undo a realized co-jump loss. Inventory-dependent quoting can nevertheless change its frequency and conditional size.

3 A finite-state model of inventory control

3.1 States, actions, and rewards

Let $\mathcal{X}$ be a finite set of states $x = (q, z)$, with $q \in \{-Q, \dots, Q\}$ and z an observable market state. The latter can encode a discretized spread, queue position, or flow regime. At state x , the maker

chooses an action $a \in \mathcal{A}_x$. The set $\mathcal{A}_x$ is compact and can be a finite union of compact quote regions, one for each choice of active sides. No action allows an inventory transition outside the inventory range.

Let $\epsilon_a = -1, \epsilon_b = 1$. A fill of side s carries a price jump j and a destination state y , with intensity measure $\nu_x^s(a, dj, \{y\})$, supported on $q(y) = q(x) + \epsilon_s$. Non-fill state transitions have rates $\ell_{xy}(a)$. The resulting off-diagonal state rates are

$$k_{xy}(a) = \ell_{xy}(a) + \sum_{s \in \{a, b\}} \int_{\mathbb{R}} \nu_x^s(a, dj, \{y\}), \quad y \neq x.$$

Transitions that leave x unchanged need not enter the generator; their price or cash effects still enter the reward.

Assume that the total predictable price drift is $dA_t = b_{X_{t-}}(a_t) dt$, including the compensators of price jumps. With running inventory penalty $\psi_x \geq 0$, define

$$r_x(a) = q(x)b_x(a) - \psi_x + \sum_s \sum_y \int_{\mathbb{R}} (\delta^s(a) + \epsilon_s j) \nu_x^s(a, dj, \{y\}). \quad (2)$$

This follows directly from (1). In particular the term qb uses the *total* price compensator, while $\epsilon_s j$ records the inventory change at a fill. If the price is instead specified using uncompensated jumps and a between-jump drift, that drift must first be converted to b .

Assumption 3.1 (Finite-state control model). For every state the rates k_{xy} are nonnegative, bounded, and continuous on each compact action region. The rewards r_x are bounded and continuous there. Marked fill laws have uniformly bounded second moments per unit time, and the remaining price martingale and drift satisfy Assumption 2.1. Terminal inventory cost Φ_x is bounded. Coefficients depend on (q, z, a) , not on the absolute price or cash.

For an admissible predictable policy π , define the remaining penalized marked-wealth value

$$J^\pi(t, x) = \mathbb{E}_{t,x}^\pi \left[\int_t^T r_{X_s}(a_s) ds - \Phi_{X_T} \right], \quad h_x(t) = \sup_\pi J^\pi(t, x).$$

The generator acts on a state vector u as

$$(L^a u)_x = \sum_{y \neq x} k_{xy}(a)(u_y - u_x), \quad \mathcal{H}_x(a; u) = r_x(a) + (L^a u)_x.$$

The cash and absolute-price coordinates have disappeared because the criterion is linear in marked wealth. This reduction does not apply unchanged to exponential utility of wealth.

3.2 Existence, verification, and the policy loss identity

Classical result 3.2 (Finite-state verification). *Under Assumption 3.1, the system*

$$\partial_t h_x(t) + \max_{a \in \mathcal{A}_x} \mathcal{H}_x(a; h(t)) = 0, \quad h_x(T) = -\Phi_x \quad (3)$$

has a unique continuously differentiable solution. It is the value function. A measurable maximizing selector is an optimal Markov policy. For every admissible policy,

$$h_x(t) - J^\pi(t, x) = \mathbb{E}_{t,x}^\pi \int_t^T \left[\max_{a \in \mathcal{A}_{X_s}} \mathcal{H}_{X_s}(a; h(s)) - \mathcal{H}_{X_s}(a_s; h(s)) \right] ds. \quad (4)$$

Proof. If the maximal total exit rate is K , then $\|L^a u - L^a v\|_\infty \leq 2K\|u - v\|_\infty$. Taking maxima preserves this Lipschitz bound. The finite-dimensional terminal-value ODE therefore has a unique solution throughout the finite horizon. Compactness and continuity give maximizers, and a measurable selector exists for the continuous Hamiltonian on a finite union of compact subsets of Euclidean space.

Apply Dynkin's formula to $h_{X_s}(s)$ under π . Using the terminal condition gives

$$J^\pi(t, x) - h_x(t) = \mathbb{E}_{t,x}^\pi \int_t^T [\partial_s h_{X_s}(s) + \mathcal{H}_{X_s}(a_s; h(s))] ds.$$

Equation (3) gives (4). Its integrand is nonnegative, and is zero under a maximizing selector. Bounded coefficients justify Dynkin's formula also for history-dependent predictable policies. $\square$

Remark 3.3 (What is optimized). The result computes inventory shadow values through (3). It does not assume them known independently of the fill law. Queue states may be included in z , but an exact exchange model requires specifying its transition rules and admissible actions. A finite-state abstraction by itself is not a formalization of an exchange rulebook.

3.3 Book events, tagged orders, and admissible states

The state must retain the variables that determine both execution priority and the maker's admissible messages. A concrete finite book cell can include the spread, external depth at finitely many ticks, the maker's posted levels, the number of unit lots ahead of each tagged order, inventory, and the status of pending messages. A missing order has a separate inactive symbol; rank zero means an active order at the front of its queue. Caps on depth and spread require declared boundary transitions or a stopping convention.

The following specification uses unit lots and strict price-time priority within a level. It is a mathematical cell, with its matching rule stated explicitly. More general priority rules require a different state and different event maps. Continuous-time queue models and state-dependent order-flow intensities are established constructions (Cont et al., 2010; Huang et al., 2015).

Event	Update at a tagged order's price
External limit addition	External depth increases; a new same-price order joins behind the tagged order and leaves its rank unchanged.
Cancellation ahead	External depth and the number of lots ahead both decrease by one.
Cancellation behind	External depth decreases; the tagged rank does not.
Market order reaching the level	When rank is positive, consume one lot ahead. At rank zero, fill the tagged unit, change inventory, and remove its active status.
Quote replacement	Remove the old tagged order and insert the new order at the back of its destination queue when the replacement is accepted.
Price or level change	Update all relative levels and ranks using a declared replenishment and relabeling map; retain any simultaneous fill-price mark.

These updates define maps $T_e(x)$ once the finite event type e includes its level, affected queue, and destination. Random replenishment can be represented by separate destination-labelled event channels. An event that cannot occur in a state has intensity zero. The maps preserve the chosen finite state space, or enter an explicit boundary state. Accepted messages and external events must use the same convention for whether displayed depth includes the maker's own order.

For a hard inventory bound, let $o_b(x), o_a(x)$ be upper bounds on the maker's remaining executable buy and sell quantities, including orders that can become live through pending messages. Admissible states and message maps preserve

$$q(x) + o_b(x) \leq Q, \quad q(x) - o_a(x) \geq -Q. \quad (5)$$

A buy fill increases q and reduces o_b by the same amount; a sell fill has the analogous property. Submission reserves the corresponding quantity, and accepted cancellation releases it. Thus the invariant prevents a live executable order from being artificially made unfillable merely because an inventory boundary has been reached. The simpler boundary convention in Assumption 2.1 corresponds to states with the inadmissible side already inactive.

The generator of an observed finite event model is

$$(L^a u)_x = \sum_e \lambda_e(x, a) [u_{T_e(x)} - u_x]. \quad (6)$$

Let $g_e(x, a)$ be the conditional mean of offset capture plus the fill-price co-jump term, less any event fee. For non-fill book events there is no offset capture or inventory-change mark. Then

$$r_x(a) = c_x(a) + \sum_e \lambda_e(x, a) g_e(x, a), \quad c_x(a) = q(x) b_x(a) - \psi_x. \quad (7)$$

Here b remains the total predictable price drift. The contribution of price jumps to carried inventory is already in qb and must not be added a second time in g_e . For an ask fill with mean price jump $\bar{j}_e$, for example, $g_e = \delta^a - \bar{j}_e - \text{fee}_e$. Equations (6)–(7) recover the rates and reward of the control model, including state-preserving events that have nonzero cash consequences.

Full-state feedback requires that this state be observable or otherwise available through a declared sufficient statistic. A hidden reserve or flow regime cannot be supplied to the policy as if observed. Such a model requires control on the posterior state; its generally continuous belief simplex is not automatically covered by a finite-state HJB.

3.4 Queue priority and competing execution times

Consider one tagged unit with n lots ahead. Opposite market orders that reach this price level arrive at rate $\mu > 0$. While $j \geq 1$ lots remain ahead, cancellations of that ahead volume occur at aggregate rate $c_j \geq 0$. Arrivals and cancellations behind do not change the rank. At rank zero the next market order fills the unit. A separate independent clock of rate $\xi \geq 0$ invalidates the quote before execution. This clock does not represent a price jump simultaneous with the fill; that jump, if present, belongs in the fill reward.

Proposition 3.4 (The value of a tagged queue position). *Let τ_n be the fill time without invalidation, and let $\sigma \sim \text{Exp}(\xi)$ be independent, with $\sigma = \infty$ if $\xi = 0$. For a discount rate $\beta \geq 0$,*

$$\mathbb{E}[e^{-\beta\tau_n} \mathbf{1}_{\{\tau_n < \sigma\}}] = \frac{\mu}{\mu + \beta + \xi} \prod_{j=1}^n \frac{\mu + c_j}{\mu + c_j + \beta + \xi}. \quad (8)$$

If $\beta + \xi > 0$, this quantity strictly decreases with the number of lots ahead. A constant positive net reward R at execution has value R times (8).

When $c_j = 0$, τ_n is Erlang with shape $n + 1$ and rate μ . In particular, for $n \geq 1$, $\mathbb{P}(\tau_n \leq t) = O(t^{n+1})$ as $t \downarrow 0$; this waiting time is not exponential.

Proof. The rank decreases successively from n to zero, with independent holding times of rates $\mu + c_n, \dots, \mu + c_1$, followed by an independent rate- μ waiting time to execution. The strong Markov property and memorylessness justify this decomposition. Independence of invalidation replaces β by $\beta + \xi$ in the Laplace transform. Multiplying the transforms of the holding times proves (8). Each extra factor is strictly between zero and one when $\beta + \xi > 0$. With no cancellations, the sum is Erlang and its distribution function is

$$1 - e^{-\mu t} \sum_{j=0}^n \frac{(\mu t)^j}{j!}.$$

Its small-time order proves the final assertion. $\square$

The conclusion concerns a fixed positive execution reward and these specified hazards. It does not assert that queue priority improves every inventory-dependent objective or every rank-dependent mark law. It does show why a constant Poisson fill intensity is insufficient for a tagged order that must first traverse a queue.

For a general finite transient queue model, let S be its subgenerator, with nonnegative off-diagonal entries and nonpositive row sums. Its missing row mass records execution or invalidation. Let f_i be the rate of execution from state i , and let $g_i = f_i R_i$, where R_i is the conditional mean execution reward, including any simultaneous mark. For a deadline H ,

$$P_i(\text{fill by } H) = \left[\int_0^H e^{tS} f dt \right]_i, \quad V_i(H) = \left[\int_0^H e^{tS} g dt \right]_i. \quad (9)$$

The entries of e^{tS} are the probabilities of occupying each transient state before absorption; multiplying by the terminal event rate and integrating proves both formulas. Discounting replaces S by $S - \beta I$. This representation retains the association between the fill state and its mark.

Corollary 3.5 (A queue-model perturbation bound). *For two valid subgenerators $S, \hat{S}$, suppose $\|S - \hat{S}\|_{\infty \rightarrow \infty} \leq \epsilon_S$, $\|g - \hat{g}\|_{\infty} \leq \epsilon_g$, and $\|g\|_{\infty} \leq G$. Then*

$$\|V(H) - \hat{V}(H)\|_{\infty} \leq H\epsilon_g + \frac{1}{2}H^2G\epsilon_S.$$

The same statement applies to fill probabilities with g replaced by f . If a finite menu of queue placements has a common uniform value bound ζ of this form, the estimated best placement has true value loss at most 2ζ .

Proof. Both substochastic semigroups contract the sup norm. Duhamel's formula bounds $\|e^{tS} - e^{t\hat{S}}\|$ by $t\epsilon_S$. Split the integrand error into a semigroup error acting on g and a reward error acted on by the estimated semigroup, then integrate. For the menu, insert the two estimated values at the true and estimated maximizing placements. $\square$

Example 3.6 (Equal displayed depth, different priority value). Four displayed lots including the tagged unit can place it either first or fourth, depending on order arrival sequence. With $\mu = 1$, no cancellations, and $\beta + \xi = 1$, the corresponding discounted execution probabilities are $1/2$ and $1/16$. Displayed depth and quote price agree, but execution value differs. No model using those two observables alone can represent both tagged states exactly.

In the same cell, a front-of-queue order paying $R = 1/2$ is worth $1/4$. Replacing it by an order paying $7/10$ with one lot ahead is worth $7/40$, or $33/200$ after a fee of $1/100$. The larger execution payoff does not compensate for priority loss. The replacement does improve on the old order with two lots ahead, whose value is $1/16$. This comparison concerns one replacement followed by holding, rather than an unrestricted switching policy.

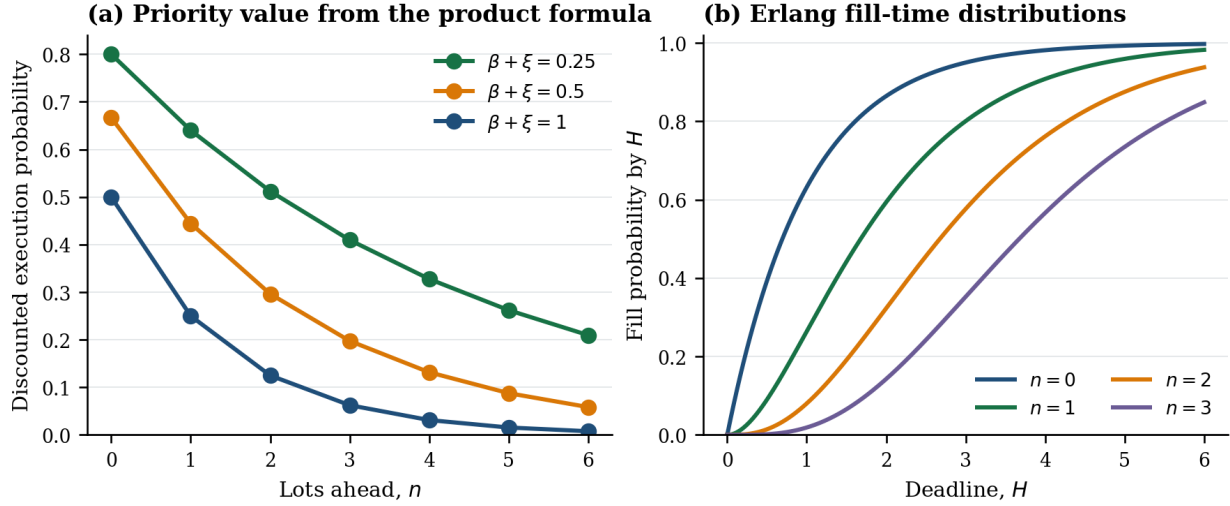


Figure 1: Queue priority in the tagged-order model. Panel (a) evaluates (8) with $\mu = 1$, no ahead cancellations, and three values of $\beta + \xi$; every extra lot ahead contributes another factor $\mu/(\mu + \beta + \xi)$. Panel (b) gives the exact Erlang fill probabilities by deadline for ranks $n = 0, 1, 2, 3$. These are analytic model evaluations, not estimates from market data.

Figure 1 displays the two analytic consequences of tagged rank used in this section: discounted priority value and the nonexponential finite-horizon fill law.

3.5 Batch consumption, partial fills, and marked volume

A unit-order absorption model does not determine the fill distribution of a larger tagged order. Let n lots be ahead of a tagged order with m lots remaining. An aggressive event consuming B lots at this price executes

$$f = \min\{m, (B - n)_+\}, \quad n' = (n - B)_+, \quad m' = m - f, \quad q' = q + sf, \quad (10)$$

where $s = +1$ for a bid and $s = -1$ for an ask. The residual order retains its priority after a partial fill in this price-time cell. An ahead cancellation removes ahead lots without executing the tagged order. Rules that requeue the residual quantity require a different state map.

For a bid, $q' + m' = q + m$; for an ask, $q' - m' = q - m$. Consequently a valid reservation remains valid after any partial fill, including a batch that exhausts the order. A pending cancel retains the unfilled reservation m' until acknowledgement. At acknowledgement only the remaining quantity is withdrawn: executed inventory is not reversed.

Let δ be the per-lot offset and J the simultaneous reference-price jump. The complete jump in marked wealth is

$$\Delta W = \delta f + (q + sf)J.$$

When carried inventory is already accounted for through its total price compensator, the event's additional fill reward is $\delta f + sfJ$. Thus its expected contribution includes $\mathbb{E}[fJ \mid x, e]$. Multiplying expected fill volume by an unconditional mean mark generally omits the dependence of price selection on the size of the consuming event.

Proposition 3.7 (A partial-fill transform). *In a finite event cell, event e at state x has rate $\lambda_e(x)$, destination $T_e(x)$, and integer executed volume $f_e(x) \geq 0$. Track a fixed tagged order, with no*

submission of additional tagged volume, and make cancellation and full execution absorbing for its fill count. Let C_t be its cumulative executed volume. For $z \geq 0$, define

$$(\mathcal{Q}_z u)_x = \sum_e \lambda_e(x) \{z^{f_e(x)} u_{T_e(x)} - u_x\}, \quad 0^0 := 1.$$

Then

$$\mathbb{E}_x[z^{C_t}] = (e^{t\mathcal{Q}_z} \mathbf{1})_x. \quad (11)$$

In particular, the coefficients give the complete partial-fill distribution, and derivatives at $z = 1$ give its factorial moments. For a bounded expected marked-reward rate $r_x = \sum_e \lambda_e(x) \mathbb{E}[g_e \mid x, e]$, the expected discounted cumulative event reward is

$$\mathbb{E}_x \left[\sum_{s \leq t} e^{-\beta s} g_s \right] = \left[\int_0^t e^{s(\mathcal{Q}_1 - \beta I)} r \, ds \right]_x. \quad (12)$$

Proof. Condition on the first infinitesimal time interval. No event leaves the transform unchanged; an event contributes the factor $z^{f_e(x)}$ and the continuation transform at $T_e(x)$. The resulting backward equation is $\partial_t u = \mathcal{Q}_z u$, $u(0) = \mathbf{1}$, proving (11). The bounded tagged volume makes its generating function a polynomial, justifying differentiation. For the reward identity, the compensation formula gives the integral of the state-dependent expected reward rate. The state transition semigroup is $e^{t\mathcal{Q}_1}$. $\square$

State-preserving events must retain their factor z^{f_e} in the tilted operator if they execute volume. At $z = 1$, the operator is the ordinary generator, and state-preserving terms cancel there as required.

Corollary 3.8 (Compound-Poisson consumption). *Suppose aggressive arrivals form a rate- λ Poisson process, their independent positive integer sizes are $B_1, B_2, \dots$, and ahead cancellations and invalidation are absent. With $S_t = \sum_{i=1}^{N_t} B_i$, a tagged order initially at (n, m) satisfies*

$$C_t = \min\{m, (S_t - n)_+\}, \quad \mathbb{P}(C_t \geq j) = \mathbb{P}(S_t \geq n + j), \quad 1 \leq j \leq m.$$

Consequently $\mathbb{E}C_t = \sum_{j=1}^m \mathbb{P}(S_t \geq n + j)$. If

$$k_* = \min\{k \geq 1 : \mathbb{P}(B_1 + \dots + B_k \geq n + 1) > 0\},$$

then, as $t \downarrow 0$,

$$\mathbb{P}(C_t > 0) = \frac{\lambda^{k_*}}{k_*!} \mathbb{P}(B_1 + \dots + B_{k_*} \geq n + 1) t^{k_*} + O(t^{k_*+1}).$$

Proof. Consumption first removes all ahead volume and then executes the tagged order, giving the pathwise identity. The tail-sum formula gives its mean. In the Poisson mixture for the first-fill probability, all counts below k_* have zero contribution. The first nonzero term has the stated coefficient, and the probability of more than k_* arrivals is $O(t^{k_*+1})$. $\square$

For example, let $n = 3, m = 2, \lambda = 1.4$, with $\mathbb{P}(B = 1) = 0.7, \mathbb{P}(B = 3) = 0.3$. At least two events are needed, and $\mathbb{P}(B_1 + B_2 \geq 4) = 0.51$. Hence

$$\mathbb{P}(C_t > 0) = \frac{2499}{5000} t^2 + O(t^3).$$

A constant positive tagged-fill hazard would instead have a linear leading term. Matching expected consumed volume $\lambda \mathbb{E}B$ does not repair this mismatch in short-latency exposure.

3.6 Cancellation and replacement at decision opportunities

A continuously adjustable offset does not by itself model the loss of time priority when an order is replaced. To keep messages within the finite-rate framework, introduce an independent decision clock of rate ν . At an opportunity, message b maps the current book state to $R_b(x)$, with immediate fee $c_x(b) \geq 0$. The finite menu contains a no-op with $R_0(x) = x, c_x(0) = 0$. All maps preserve (5). Between decision opportunities the posted quote and tagged rank are state variables.

Proposition 3.9 (A verified quote-replacement model). *Let L^0, r^0 describe bounded external book events and their marked reward. The finite-horizon value satisfies*

$$\partial_t h_x + r_x^0 + (L^0 h)_x + \nu \max_b \{h_{R_b(x)} - h_x - c_x(b)\} = 0, \quad h_x(T) = -\Phi_x. \quad (13)$$

It has the verification and performance-difference identity of Classical result 3.2. In particular, a proposed replacement is compared with keeping the existing order through its entire post-message continuation value, including its new rank.

Proof. For a fixed message choice, the clock contributes generator $\nu[u_{R_b(x)} - u_x]$ and expected fee rate $-\nu c_x(b)$. Thus its Hamiltonian is $r_x^0 + (L^0 u)_x + \nu[u_{R_b(x)} - u_x - c_x(b)]$. The finite action menu and bounded coefficients satisfy the hypotheses of Classical result 3.2. Taking the maximum over the menu gives (13), and the same Dynkin argument gives the performance identity. $\square$

An accepted atomic replacement is the message convention in this model. If cancellation and insertion have separate delays, the state must include the pending stages and permit fills of the still-live order. Finite exponential stages provide another finite-state model; a deterministic delay generally requires an additional age variable. Neither model allows free continuous relocation while retaining the old queue position. Discrete message choices use value-gap or hybrid certificates; smooth offset curvature is not a property of a tick grid.

3.7 Live orders during cancellation latency

A decision opportunity and an exchange acknowledgement are different times. Latency and adverse selection in limit-order control are already studied by [Lehalle and Mounjid \(2017\)](#). The comparison here concerns the error made when a specified pending-message delay is replaced by an instantaneous reset or by finitely many exponential stages.

After sending a cancellation, let X be the finite state of the still-live book and tagged order, with generator Q , until acknowledgement. The state includes remaining quantity and all inventory changes from partial fills. Further messages are suspended during this pending period. Let r be its expected marked-reward rate, including running inventory cost, and let w_x be the value after acknowledging the cancellation from pre-acknowledgement state x . Thus w withdraws only the residual order and retains the inventory actually accumulated. The acknowledgement delay $D \geq 0$ has a common law across initial states, is independent of future book events, and has finite mean. No independence of fill size and price mark is required.

Theorem 3.10 (The value and perturbation of a pending cancellation). *For $\beta \geq 0$, put $A = Q - \beta I$ and $g = r + Aw$. The value of the pending period and its continuation is*

$$V^D = \mathbb{E} \left[\int_0^D e^{tA} r \, dt + e^{DA} w \right], \quad (14)$$

$$V^D - w = \mathbb{E} \int_0^D e^{tA} g \, dt. \quad (15)$$

In particular, with $G = \|g\|_\infty$,

$$\|V^D - w\|_\infty \leq G \mathbb{E} \int_0^D e^{-\beta t} dt \leq G \mathbb{E} D.$$

For two independent-delay laws with finite first moments,

$$\|V^D - V^{D'}\|_\infty \leq G W_1(\mathcal{L}(D), \mathcal{L}(D')), \quad (16)$$

where W_1 is the minimum of $\mathbb{E}|D - D'|$ over couplings of their marginal delay laws.

Proof. Condition on the delay. The finite-state reward and terminal-value formulas give (14). Differentiation in a deterministic delay t yields

$$\frac{d}{dt} \left\{ \int_0^t e^{sA} r ds + e^{tA} w \right\} = e^{tA} (r + Aw) = e^{tA} g.$$

Integrating from zero proves (15). The discounted Markov semigroup satisfies $\|e^{tA}\|_{\infty \rightarrow \infty} \leq e^{-\beta t}$. This proves the first bound and makes the deterministic-delay value G -Lipschitz in time. Apply that Lipschitz inequality to any coupling of D, D' , take expectations, and then minimize over couplings. $\square$

The sign of g is not fixed. Delay can admit a profitable fill or a damaging fill, change the inventory penalty, and discount the post-acknowledgement continuation. A latency allowance must retain those channels. If congestion makes the delay dependent on future order flow, its marginal distribution does not determine (14); the joint state and acknowledgement law must be modeled.

Theorem 3.11 (A sharp finite-stage approximation of deterministic latency). *In Theorem 3.10, fix $d > 0$ and replace the deterministic delay d by an independent Erlang delay D_k with k stages of rate $\kappa = k/d$. Let*

$$B = \|Ag\|_\infty, \quad C = \|A^2g\|_\infty.$$

The finite-stage value is computed by

$$v_0 = w, \quad [(\kappa + \beta)I - Q]v_j = r + \kappa v_{j-1}, \quad j = 1, \dots, k, \quad V^{D_k} = v_k. \quad (17)$$

It satisfies the uniform bounds

$$\|V^{D_k} - V^d\|_\infty \leq \min \left\{ \frac{Gd}{\sqrt{k}}, \frac{Bd^2}{2k} \right\}, \quad (18)$$

and the explicit expansion

$$\left\| V^{D_k} - V^d - \frac{d^2}{2k} e^{dA} Ag \right\|_\infty \leq \frac{Cd^3}{6k^{3/2}} \sqrt{3 + \frac{6}{k}}. \quad (19)$$

The order k^{-1} in (18) cannot be improved uniformly over finite queue cells.

Proof. Conditioning on the first exponential stage gives the resolvent equation (17). Its inverse exists because $\kappa + \beta > 0$ and Q is a finite generator. The Erlang law has mean d and variance d^2/k . The Wasserstein bound against the constant d , followed by Cauchy–Schwarz, gives the first term in (18).

Write the deterministic-delay value as $f(t)$. The preceding proof gives $f'(t) = e^{tA}g$, $f''(t) = e^{tA}Ag$, and $f'''(t) = e^{tA}A^2g$. Their sup-norm bounds are respectively G, B, C . Taylor expansion about d , with $\mathbb{E}(D_k - d) = 0$, gives the second bound. Keeping the second-order term gives the leading vector in (19). The absolute third-order remainder is bounded using

$$\begin{aligned}\mathbb{E}|D_k - d|^3 &\leq \{\mathbb{E}(D_k - d)^2 \mathbb{E}(D_k - d)^4\}^{1/2}, \\ \mathbb{E}(D_k - d)^4 &= \frac{3d^4}{k^2} + \frac{6d^4}{k^3}.\end{aligned}$$

These identities give the displayed remainder.

For sharpness, place one tagged unit at the front of a queue, with fill rate $\mu > 0$, positive fill reward R , no other reward, and zero post-cancellation value. Until acknowledgement the unit remains executable. Then

$$V^d = \frac{\mu R}{\mu + \beta}(1 - e^{-(\mu + \beta)d}), \quad V^{D_k} = \frac{\mu R}{\mu + \beta}\{1 - (1 + (\mu + \beta)d/k)^{-k}\}.$$

Expanding the logarithm shows

$$k(V^{D_k} - V^d) \longrightarrow -\frac{\mu R(\mu + \beta)d^2}{2}e^{-(\mu + \beta)d} \neq 0.$$

Thus a genuine tagged-fill example attains the stated order. $\square$

The first bound only uses control of the random time displacement. Matching the mean cancels its linear contribution to the expected value, which explains the sharper k^{-1} rate. This is a bound on value approximation, not on pathwise proximity of acknowledgement times. Known stage-approximation allowances can be added to local message-value intervals. For a finite menu of one-time message choices with uniform error ζ , choosing the best approximate value loses at most 2ζ , by the same two-value comparison as Corollary 3.5.

3.8 From event-law errors to Hamiltonian error budgets

The event representation makes the primitive bounds used below traceable to the book model. Fix a smooth action branch on which the event maps $T_e(x)$ agree in the true and estimated models. If the destination of an event is uncertain, index channels by both event type and destination and put the uncertainty in their rates. Moving a destination without recording that change in the rates is outside this comparison.

Proposition 3.12 (Event-law derivative budgets). *Suppose the rates, conditional mean rewards, and base rewards in (7) have derivatives through order two on the branch. For $j = 0, 1, 2$, let uniform bounds satisfy*

$$\begin{aligned}\sum_e \|D^j(\lambda_e - \hat{\lambda}_e)\| &\leq \varepsilon_{\lambda,j}, & \sup_e \|D^j(g_e - \hat{g}_e)\| &\leq \varepsilon_{g,j}, \\ \|D^j(c - \hat{c})\| &\leq \varepsilon_{c,j}, & \sup_e \|D^j g_e\| &\leq G_j, \\ \sum_e \|D^j \hat{\lambda}_e\| &\leq \hat{L}_{\lambda,j}.\end{aligned}$$

Here order zero means absolute value and higher orders use the corresponding operator norms. The off-diagonal transition rates $k_{xy} = \sum_{e: T_e(x)=y} \lambda_e$, $y \neq x$, and running rewards obey

$$\sum_{y \neq x} \|D^j(k_{xy} - \hat{k}_{xy})\| \leq \varepsilon_{\lambda,j}, \quad (20)$$

$$\|D^j(r - \hat{r})\| \leq \varepsilon_{c,j} + \sum_{\ell=0}^j \binom{j}{\ell} (G_{j-\ell} \varepsilon_{\lambda,\ell} + \hat{L}_{\lambda,\ell} \varepsilon_{g,j-\ell}). \quad (21)$$

Proof. Group the event rates by their common destination and apply the triangle inequality. Omitting self-events cannot increase the bound in (20). For the reward, write

$$r - \hat{r} = c - \hat{c} + \sum_e [(\lambda_e - \hat{\lambda}_e)g_e + \hat{\lambda}_e(g_e - \hat{g}_e)].$$

Apply the product rule. At order two, the two cross terms are outer products whose operator norms are bounded by the products of the gradient norms. Taking norms and using the displayed bounds gives (21) at each order. $\square$

Order zero supplies the value-error budget. Order one supplies the gradient discrepancy, and order two supplies the curvature enclosure in Section 7. The quantities G_j can themselves be bounded by fitted derivatives plus the corresponding uncertainty bound. These are assumptions or certified enclosures, not consequences of observing a finite sample. The g_e are conditional means; the proposition does not assume that every realized price mark is bounded. A derivative certificate applies within a fixed smooth branch. Queue resets, participation changes, and ticks require the finite-action or hybrid comparison below.

3.9 The cost of truncating the book

A finite depth cap changes the model when the book reaches its boundary. The following comparison separates that error from parameter estimation and numerical solution.

Proposition 3.13 (A coupling certificate for a finite cell). *Suppose a nonexplosive full book and a capped book can be coupled to have identical states, feasible actions, and rewards up to the first exit time τ from their common retained cell. Policies on the two models correspond before exit and admit feasible extensions after exit. Terminal payoffs agree within the retained cell. Assume $|r| \leq R_\infty$ and $|\Phi| \leq \Phi_\infty$ in both models. For any corresponding policy pair, started at (t, x) inside the cell,*

$$|J_{\text{full}}^\pi - J_{\text{cap}}^\pi| \leq 2\mathbb{E}[\mathbf{1}_{\{\tau \leq T\}} \{\Phi_\infty + R_\infty(T - \tau)\}] \leq 2B(t) \mathbb{P}(\tau \leq T), \quad (22)$$

where $B(t) = \Phi_\infty + R_\infty(T - t)$. If the exit probability is at most p uniformly over the compared policy class, then

$$|V_{\text{full}} - V_{\text{cap}}| \leq 2B(t)p. \quad (23)$$

A policy with regret at most C in the capped model therefore has regret at most $C + 4B(t)p$ in the full model.

Proof. The pathwise reduced rewards coincide before exit, and the complete payoffs coincide on $\{\tau > T\}$. On exit, the difference between the remaining integrated rewards is at most $2R_\infty(T - \tau)$, and the terminal difference is at most $2\Phi_\infty$. Taking expectations gives (22). Uniformity and policy correspondence permit taking suprema, which proves (23). Finally, insert the capped optimal value and the capped deployed-policy value between the corresponding full-model values. The two comparison errors are each at most $2B(t)p$, and the middle gap is at most C . $\square$

The uniform exit bound must cover the deployed policy and the policies against which it is compared. A small boundary frequency under a logging policy alone does not establish it. Reward bounds and feasible post-exit extensions are also substantive assumptions. This result allows the finite-cell certificates to be used with an explicit truncation allowance; it does not justify an arbitrary replenishment rule from the fact that the retained state space is finite.

4 Offsets, inventory shadow values, and selection effects

Consider one active side at a fixed state and time. Suppose its Hamiltonian contribution can be written

$$R(\delta) = (\delta - p)\Lambda(\delta),$$

where $p \in \mathbb{R}$ is independent of the offset. For a fill leaving the market state unchanged and having conditional mean jump m_s ,

$$p = h_{q,z}(t) - h_{q+\epsilon_s,z}(t) - \epsilon_s m_s.$$

An inventory-reducing fill can have negative shadow cost, so p is not restricted to be nonnegative.

Proposition 4.1 (The quote equation). *Let $\Lambda > 0$ be continuously differentiable and decreasing. At an interior maximizer with $\Lambda'(\delta^*) < 0$,*

$$\delta^* = p - \frac{\Lambda(\delta^*)}{\Lambda'(\delta^*)}.$$

For $\Lambda(\delta) = \Lambda_0 e^{-\kappa\delta}$, the maximizer on $[0, D]$ is

$$\delta^* = \min\{D, \max\{0, p + 1/\kappa\}\}.$$

If withdrawal is allowed, compare the resulting Hamiltonian contribution with the value zero of an inactive side.

Proof. Differentiate R . For the exponential law, $R' = \Lambda_0 e^{-\kappa\delta}[1 - \kappa(\delta - p)]$, whose sign changes at most once, from positive to negative. Clipping its unconstrained maximizer to the interval proves the assertion. Withdrawal is a separate feasible action. $\square$

The exponential objective is not globally concave:

$$R''(\delta) = \Lambda_0 e^{-\kappa\delta}[\kappa^2(\delta - p) - 2\kappa].$$

Strong-concavity certificates below apply only on regions on which this second derivative has a uniform negative upper bound. Also, the usual demand elasticity is $-\delta\Lambda'/\Lambda = \kappa\delta$, which is not constant. The constant object is the inverse semi-elasticity $-\Lambda/\Lambda' = 1/\kappa$.

Proposition 4.2 (Offset-dependent adverse selection). *If the expected mark or continuation value conditional on a fill depends on the offset, write $R(\delta) = (\delta - p(\delta))\Lambda(\delta)$. For differentiable p , an interior optimum satisfies*

$$0 = (1 - p'(\delta^*))\Lambda(\delta^*) + (\delta^* - p(\delta^*))\Lambda'(\delta^*).$$

Thus a fixed markup plus a fixed shadow cost is valid only when the relevant conditional penalty is offset-independent.

Proof. This is the product rule applied to R . $\square$

In the exponential-utility model of [Avellaneda and Stoikov \(2008\)](#), the intensity contribution is $\gamma^{-1} \log(1 + \gamma/\kappa)$, approaching $1/\kappa$ as $\gamma \downarrow 0$. The exact utility objective matters; the risk-neutral Hamiltonian above does not make all risk-aversion specifications equivalent.

5 An exact gross-profit shutdown threshold

Definition 5.1 (Two populations). On one ask side, uninformed fills arrive at rate $\lambda_u e^{-\kappa\delta}$, with $\lambda_u, \kappa > 0$, and have zero expected per-fill markout. Informed opportunities arrive at rate $\lambda_i \geq 0$. Their upward mark J is exponential with parameter $k > 0$; a trader fills the ask precisely when $J > \delta$. The objective in this section is the sum of incremental per-fill markouts, excluding pre-existing inventory, fees, and continuation value.

Memorylessness gives $\mathbb{E}[J - \delta \mid J > \delta] = 1/k$. Consequently the per-time gross markout is

$$g(\delta) = \lambda_u \delta e^{-\kappa\delta} - \frac{\lambda_i}{k} e^{-k\delta}. \quad (24)$$

This is a static component of a quoting objective; it is not a theorem that the entire dynamic inventory book should be shut down.

Proposition 5.2 (Unbounded-offset gross-markout threshold). *For the objective (24) on $\delta \geq 0$:*

1. *If $k \geq \kappa$, some finite offset has strictly positive gross profit, for every finite λ_i .*
2. *If $k < \kappa$, every offset has nonpositive gross profit if and only if*

$$\lambda_i \geq \lambda_i^* := \frac{k\lambda_u}{e(\kappa - k)}.$$

At equality the only zero at finite offset is $\delta = 1/(\kappa - k)$.

Proof. Multiplication by $e^{k\delta}$ shows that $g(\delta) > 0$ is equivalent to $\lambda_u \delta e^{-(\kappa-k)\delta} > \lambda_i/k$. For $k \geq \kappa$ the left side is unbounded. For $k < \kappa$ its unique maximum is $\lambda_u/[e(\kappa - k)]$, at the stated offset. $\square$

Corollary 5.3 (A finite quote range). *On $[0, D]$, $D > 0$, a positive quote exists if and only if*

$$\lambda_i < k\lambda_u \max_{0 \leq \delta \leq D} \delta e^{-(\kappa-k)\delta}.$$

The maximizing offset in this expression is $\min\{D, 1/(\kappa - k)\}$ if $k < \kappa$, and D if $k \geq \kappa$.

Proof. Apply the same inequality on the compact feasible interval. $\square$

Heavier informed tails correspond to $k < \kappa$, precisely the regime where the unbounded-offset shutdown threshold exists. On a finite quote range there is a threshold in both regimes. At a hard inventory limit, withdrawal may be mandatory independently of these profitability comparisons. A fill that reduces a sufficiently costly inventory can also have positive continuation value despite a negative immediate markout.

Example 5.4 (Threshold arithmetic). With $\lambda_u = 10, \kappa = 2, k = 1$, the threshold is $10/e \approx 3.678794$, attained at $\delta = 1$. At $\lambda_i = 10/e$, the two terms of $g(1)$ both equal $10e^{-2}$. Values $\lambda_i < 10/e$ leave a profitable interval; values above it make all finite quotes unprofitable for this static criterion.

Figure 2 makes the phase boundary in Proposition 5.2 explicit.

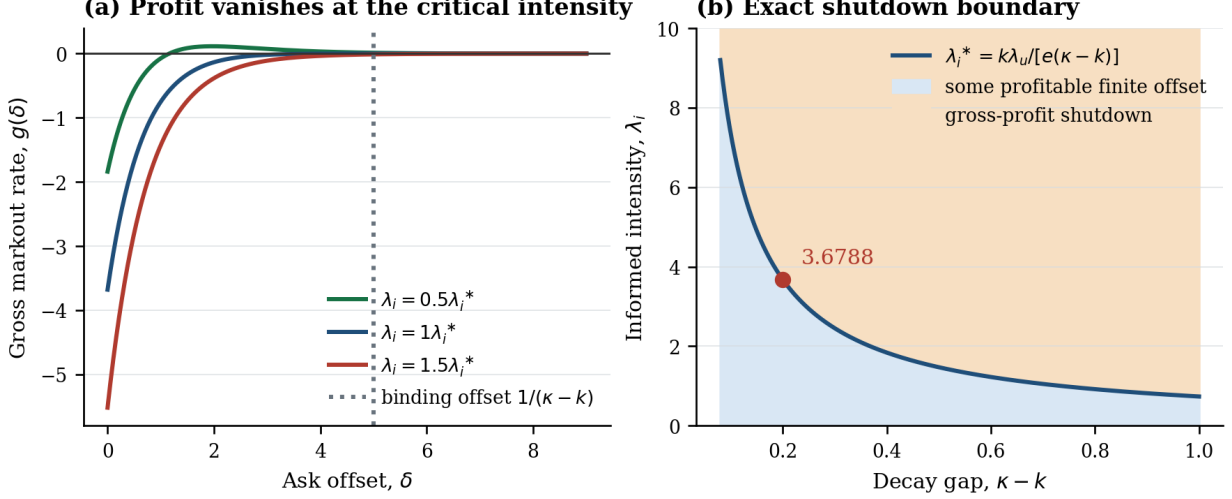


Figure 2: Exact gross-markout geometry. Panel (a) plots $g(\delta)$ for $\lambda_u = 2$, $\kappa = 1.2$, $k = 1$, and informed intensities equal to one half, exactly, and one and a half times $\lambda_i^* = 10/e$; at equality the maximum is zero at $\delta = 1/(\kappa - k) = 5$. Panel (b) plots $\lambda_i^* = k\lambda_u/[e(\kappa - k)]$ for $\lambda_u = 2$, $k = 1$. Its red point, at $\kappa - k = 0.2$, has $\lambda_i^* = 10/e$, which coincides numerically with the threshold of Example 5.4 (there $\lambda_u = 10$, $\kappa - k = 1$). Shading indicates whether some finite offset has positive gross markout under the static criterion of Proposition 5.2.

6 Certificates for a misspecified quoting model

6.1 A constrained local certificate

Lemma 6.1 (Gradient error on a convex action set). *Let $A \subset \mathbb{R}^d$ be convex and compact, and let $F, \hat{F}$ be differentiable on a neighborhood of A . Assume F is m -strongly concave on A , $m > 0$, and $a^* \in \arg \max_A F$, $\hat{a} \in \arg \max_A \hat{F}$. Then, including boundary maximizers,*

$$0 \leq F(a^*) - F(\hat{a}) \leq \frac{\|\nabla(F - \hat{F})(\hat{a})\|^2}{2m}, \quad (25)$$

$$\|a^* - \hat{a}\| \leq \frac{\|\nabla(F - \hat{F})(\hat{a})\|}{m}. \quad (26)$$

Proof. Put $d = a^* - \hat{a}$. The first-order necessary inequality for the constrained maximum of $\hat{F}$ gives $\nabla \hat{F}(\hat{a}) \cdot d \leq 0$. Strong concavity therefore gives

$$F(a^*) - F(\hat{a}) \leq \nabla(F - \hat{F})(\hat{a}) \cdot d - \frac{m}{2}\|d\|^2 \leq \frac{\|\nabla(F - \hat{F})(\hat{a})\|^2}{2m}.$$

For the distance bound, the optimality inequality for F at a^* and strong monotonicity of $-\nabla F$ imply

$$m\|d\|^2 \leq (\nabla F(\hat{a}) - \nabla F(a^*)) \cdot d \leq \nabla(F - \hat{F})(\hat{a}) \cdot d.$$

Cauchy–Schwarz completes the proof. $\square$

For a one-dimensional fixed- p reward $R = (\delta - p)\Lambda$, the residual is

$$(R - \hat{R})' = (\Lambda - \hat{\Lambda}) + (\delta - p)(\Lambda' - \hat{\Lambda}').$$

A constant multiplicative intensity error leaves a fixed- p optimizer unchanged. It need not leave the dynamic optimizer unchanged, because the intensity scale changes the time available to adjust inventory and therefore changes the value function.

6.2 From a quote error to total policy regret

Let $(\hat{r}, \hat{k})$ be a second finite-state control model with the same state set, action sets, and terminal penalty. Its value and Hamiltonian are $\hat{h}, \hat{\mathcal{H}}$. Let $\hat{\pi}(t, x)$ maximize $\hat{\mathcal{H}}_x(a; \hat{h}(t))$. Define

$$e_x(t, a) = \mathcal{H}_x(a; h(t)) - \hat{\mathcal{H}}_x(a; \hat{h}(t)).$$

This error includes the difference in continuation values. Holding h fixed while replacing the fill law would omit a policy-relevant part of the misspecification.

Theorem 6.2 (Dynamic gradient certificate). *Suppose each $\mathcal{A}_x$ is a common convex compact quote region, the two Hamiltonians are differentiable in a , and $\mathcal{H}_x(\cdot; h(s))$ is $m_x(s)$ -strongly concave with $m_x(s) \geq m_0 > 0$. Then*

$$0 \leq h_x(t) - J^{\hat{\pi}}(t, x) \leq \mathbb{E}_{t,x}^{\hat{\pi}, \text{true}} \int_t^T \frac{\|\nabla_a e_{X_s}(s, \hat{\pi}(s, X_s))\|^2}{2m_{X_s}(s)} ds. \quad (27)$$

In particular a uniform bound $\|\nabla_a e_x(s, a)\| \leq b(s)$ gives regret at most $\int_t^T b(s)^2 / (2m_0) ds$.

Proof. Use Lemma 6.1 at each time and state with $F = \mathcal{H}_x(\cdot; h(s))$ and $\hat{F} = \hat{\mathcal{H}}_x(\cdot; \hat{h}(s))$. Substitute its gap bound in (4). The expectation is under the true controlled process, exactly as in the performance identity; no equality of visitation laws is assumed. $\square$

6.3 Primitive errors and the continuation-value contribution

Let $\Phi_\infty = \max_x |\Phi_x|$, and assume both models' rewards are bounded by R_∞ . Put

$$B(s) = \Phi_\infty + (T - s)R_\infty.$$

Both value functions have sup norm at most $B(s)$. For simplicity write time-independent error budgets

$$\begin{aligned} \varepsilon_r &= \sup_{x,a} |r_x(a) - \hat{r}_x(a)|, \\ \varepsilon_k &= \sup_{x,a} \sum_{y \neq x} |k_{xy}(a) - \hat{k}_{xy}(a)|, \\ \varepsilon_{r,1} &= \sup_{x,a} \|\nabla r_x(a) - \nabla \hat{r}_x(a)\|, \\ \varepsilon_{k,1} &= \sup_{x,a} \sum_{y \neq x} \|\nabla k_{xy}(a) - \nabla \hat{k}_{xy}(a)\|, \\ L_{k,1} &= \sup_{x,a} \sum_{y \neq x} \|\nabla \hat{k}_{xy}(a)\|. \end{aligned}$$

The gradients are taken within the common convex regions.

Lemma 6.3 (Value-function perturbation). *Under the finite-state assumptions,*

$$\|h(t) - \hat{h}(t)\|_\infty \leq d(t) := \int_t^T [\varepsilon_r + 2B(s)\varepsilon_k] ds. \quad (28)$$

This estimate does not require concavity.

Proof. For $u = \hat{h}(s)$, the Hamiltonians at the same u differ uniformly by at most $c(s) = \varepsilon_r + 2B(s)\varepsilon_k$. Adding a constant vector does not change any $L^a u$. Consequently $\hat{h}(s) + d(s)\mathbf{1}$ has terminal value $-\Phi$ and satisfies $\partial_s u + \max_a \mathcal{H}(a; u) \leq 0$. Dynkin's formula shows that it dominates every true-policy value. The vector $\hat{h}(s) - d(s)\mathbf{1}$ satisfies the reverse inequality; a true-Hamiltonian maximizing selector for that vector attains a value at least as large. This brackets h between the two vectors and proves the bound. $\square$

Corollary 6.4 (A certificate in primitive model errors). *Under Theorem 6.2 and the displayed budgets, one may take*

$$b(s) = \varepsilon_{r,1} + 2B(s)\varepsilon_{k,1} + 2d(s)L_{k,1}$$

in (27). If the four error budgets are $O(\varepsilon)$ and the horizon, reward bounds, derivative bounds, and m_0^{-1} remain bounded, the policy regret is $O(\varepsilon^2)$.

Proof. Expand the gradient of e_x as

$$\begin{aligned} \nabla e_x &= \nabla(r_x - \hat{r}_x) + \sum_{y \neq x} \nabla(k_{xy} - \hat{k}_{xy})(h_y - h_x) \\ &\quad + \sum_{y \neq x} \nabla \hat{k}_{xy} [(h_y - \hat{h}_y) - (h_x - \hat{h}_x)]. \end{aligned}$$

The differences in the first sum have magnitude at most $2B(s)$; those in the second at most $2d(s)$, by Lemma 6.3. The triangle inequality gives the bound on b . The rate statement follows with a fixed positive concavity modulus. $\square$

The budgets must be justified on the region used by the policy. A fitted kernel and a fitted Hessian alone do not certify their errors relative to the true market. For random estimates, simultaneous high-probability bounds on these quantities give a certificate on the same event. Constants deteriorate near flat Hamiltonians and near an uncontrolled increase of the horizon.

6.4 Withdrawal, ticks, and branch changes

Proposition 6.5 (A value-error certificate without concavity). *On any compact action set, possibly disconnected or finite, suppose*

$$\sup_{a \in \mathcal{A}_x} |\mathcal{H}_x(a; h(s)) - \hat{\mathcal{H}}_x(a; \hat{h}(s))| \leq \zeta_x(s).$$

The estimated maximizer has true Hamiltonian gap at most $2\zeta_x(s)$. Therefore its dynamic regret is at most $\mathbb{E}^{\hat{\pi}, \text{true}} \int_t^T 2\zeta_{X_s}(s) ds$.

Proof. Let a^* maximize the true Hamiltonian. Insert the estimated Hamiltonian at a^* and at $\hat{a}$. Its optimality gives $\hat{\mathcal{H}}(a^*) \leq \hat{\mathcal{H}}(\hat{a})$, leaving at most two uniform errors. Apply (4). $\square$

If action branches represent different active sides, a true gap strictly larger than $2\zeta_x(s)$ between the best branch and every other branch preserves the selected branch. A local quadratic certificate then applies if that branch is convex and strongly concave. Without such a margin, branch switching prevents a universal quadratic bound based only on a within-branch gradient. Tick-constrained quotes are governed by the value-error bound unless an additional discrete decision margin is established.

6.5 Stationary discounting and the effective horizon

Suppose the primitives are stationary and satisfy Assumption 3.1; there is no terminal penalty. For a discount rate $\beta > 0$, define

$$J_\beta^\pi(x) = \mathbb{E}_x^\pi \int_0^\infty e^{-\beta s} r_{X_s}(a_s) ds, \quad h_x = \sup_\pi J_\beta^\pi(x).$$

Take a second stationary model with the same state and action sets, both rewards bounded by R_∞ , and the primitive budgets above. Let $\bar{K}$ bound the total exit rates of both models, and put

$$\begin{aligned} B_\beta &= R_\infty/\beta, & d_\beta &= (\varepsilon_r + 2B_\beta \varepsilon_k)/\beta, \\ b_\beta &= \varepsilon_{r,1} + 2B_\beta \varepsilon_{k,1} + 2d_\beta L_{k,1}. \end{aligned} \tag{29}$$

Theorem 6.6 (Discounted policy certificate). *The equation $\beta h_x = \max_{a \in \mathcal{A}_x} \mathcal{H}_x(a; h)$ has a unique solution. It is the discounted value, and a maximizing stationary selector is optimal. Every admissible policy satisfies*

$$h_x - J_\beta^\pi(x) = \mathbb{E}_x^\pi \int_0^\infty e^{-\beta s} \left[\max_{a \in \mathcal{A}_{X_s}} \mathcal{H}_{X_s}(a; h) - \mathcal{H}_{X_s}(a_s; h) \right] ds. \tag{30}$$

Moreover $\|h\|_\infty, \|\hat{h}\|_\infty \leq B_\beta$ and $\|h - \hat{h}\|_\infty \leq d_\beta$. If the common action sets are convex and compact, both Hamiltonians are differentiable in the actions, and every true $\mathcal{H}_x(\cdot; h)$ is m_0 -strongly concave, $m_0 > 0$, then the stationary estimated optimal policy obeys

$$0 \leq h_x - \hat{J}_\beta^\pi(x) \leq \frac{b_\beta^2}{2m_0\beta} \quad \text{for every } x. \tag{31}$$

Proof. Write $k_x(a) = \sum_{y \neq x} k_{xy}(a)$. The uniformized operator

$$(\mathcal{T}u)_x = \frac{1}{\beta + \bar{K}} \max_{a \in \mathcal{A}_x} \left[r_x(a) + \sum_{y \neq x} k_{xy}(a) u_y + (\bar{K} - k_x(a)) u_x \right]$$

is a sup-norm contraction with factor $\bar{K}/(\beta + \bar{K}) < 1$: its coefficients on u are nonnegative and sum to $\bar{K}$. Its fixed-point equation is the displayed discounted HJB equation. This proves existence and uniqueness. Discounted Dynkin's formula for the bounded vector h , followed by passage to the infinite horizon, gives

$$h_x = J_\beta^\pi(x) + \mathbb{E}_x^\pi \int_0^\infty e^{-\beta s} [\beta h_{X_s} - \mathcal{H}_{X_s}(a_s; h)] ds.$$

The transversality term vanishes by boundedness. The bracket is nonnegative and is zero under a maximizing selector, proving the value assertion and (30).

The reward bound gives $\|h\|_\infty, \|\hat{h}\|_\infty \leq B_\beta$. At the common vector $\hat{h}$, the two Hamiltonians differ by at most $c_\beta = \varepsilon_r + 2B_\beta \varepsilon_k$. Because generators annihilate constants, $u^+ = \hat{h} + d_\beta \mathbf{1}$ satisfies $\beta u_x^+ \geq \max_a \mathcal{H}_x(a; u^+)$; discounted Dynkin's formula shows $u^+ \geq h$. The reverse inequality holds for $u^- = \hat{h} - d_\beta \mathbf{1}$. Under a selector maximizing $\mathcal{H}(\cdot; u^-)$, the same formula gives $u^- \leq J_\beta^\pi \leq h$. This proves the perturbation bound. The gradient expansion in Corollary 6.4, with B_β, d_β , bounds the full gradient error by b_β . Lemma 6.1 bounds the bracket in (30) by $b_\beta^2/(2m_0)$. Integration of the discount factor proves (31). $\square$

Remark 6.7 (Dependence on the discount rate). The certificate has no terminal-horizon parameter, but its constants are not uniform as $\beta \downarrow 0$. In fact

$$b_\beta = \varepsilon_{r,1} + \frac{2R_\infty \varepsilon_{k,1} + 2L_{k,1} \varepsilon_r}{\beta} + \frac{4R_\infty L_{k,1} \varepsilon_k}{\beta^2}.$$

If all four error budgets are $O(\varepsilon)$, with uniform $R_\infty, L_{k,1}, m_0^{-1}$, the displayed bound is in general $O(\varepsilon^2 \beta^{-5})$ as $\beta \downarrow 0$, and its discount-normalized version is $O(\varepsilon^2 \beta^{-4})$. These are growth rates of this upper bound, not matching lower bounds on actual regret. The reward bound also gives $h_x - J_\beta^\pi(x) \leq 2R_\infty/\beta$, so the minimum with that bound may always be taken. If the β^{-2} contribution to b_β vanishes, for example when $\varepsilon_k = 0$, the corresponding bounds improve to $O(\varepsilon^2 \beta^{-3})$ and $O(\varepsilon^2 \beta^{-2})$. An ergodic limit requires additional stability and relative-value analysis; it does not follow by setting $\beta = 0$ here. The ergodic analysis of [Cao et al. \(2026\)](#) addresses a different regime.

7 A certificate from computed values and supplied error bounds

An exact estimated HJB solution is not a numerical output. A solver returns an approximation, and a quoting routine can stop before reaching its maximum. The following result keeps both errors in the certificate. It also verifies curvature against model uncertainty.

Suppose the common action sets are compact and convex, and the primitive rewards and rates are twice continuously differentiable on their neighborhoods. Let $v : [0, T] \rightarrow \mathbb{R}^{|\mathcal{X}|}$ be a computed continuation vector of class C^1 . Assume supplied, simultaneous bounds, uniform over states and actions, for $j = 0, 1, 2$:

$$\|\nabla^j(r - \hat{r})\| \leq \epsilon_{r,j}, \quad \sum_{y \neq x} \|\nabla^j(k_{xy} - \hat{k}_{xy})\| \leq \epsilon_{k,j}.$$

For $j = 0$ the norm is absolute value; for $j = 2$ it is the matrix operator norm. Bounds may depend on time. Let

$$\bar{K}_j(s) \geq \sup_{x,a} \sum_{y \neq x} \|\nabla^j \hat{k}_{xy}(a)\| + \epsilon_{k,j}(s), \quad j = 1, 2.$$

Thus $\bar{K}_j$ also bounds the true rate derivatives. Define the computed HJB and terminal residual bounds by

$$\rho(s) \geq \max_x \left| \partial_s v_x(s) + \max_a \hat{\mathcal{H}}_x(a; v(s)) \right|, \quad d_T \geq \|v(T) + \Phi\|_\infty,$$

and write $\text{span}(v) = \max_x v_x - \min_x v_x$. Set

$$\begin{aligned} c(s) &= \epsilon_{r,0}(s) + \text{span}(v(s)) \epsilon_{k,0}(s), \\ D(t) &= d_T + \int_t^T [\rho(s) + c(s)] ds, \\ b_j(s) &= \epsilon_{r,j}(s) + \text{span}(v(s)) \epsilon_{k,j}(s) + 2D(s) \bar{K}_j(s), \quad j = 1, 2. \end{aligned} \tag{32}$$

For $a \in A$, use the outward normal cone

$$N_A(a) = \{n : n^\top(z - a) \leq 0 \text{ for every } z \in A\}.$$

For a computed Markov quote $\tilde{a} = \tilde{\pi}(s, x)$, its first-order optimization residual is

$$\chi_x(s) = \text{dist}(\nabla_a \hat{\mathcal{H}}_x(\tilde{a}; v(s)), N_{A_x}(\tilde{a})).$$

It is zero at an exact constrained maximizer; it also accommodates boundary quotes without imposing an interior stationarity equation.

Theorem 7.1 (Computable model-and-solver certificate). *Under the finite-state assumptions and the displayed bounds, $\|h(t) - v(t)\|_\infty \leq D(t)$. Suppose a verified lower curvature bound satisfies*

$$-\nabla_a^2 \hat{\mathcal{H}}_x(a; v(s)) \succeq \hat{m}_x(s)I \quad \text{on } \mathcal{A}_x, \quad m_x^-(s) := \hat{m}_x(s) - b_2(s) > 0.$$

Assume the reciprocal lower moduli and the bounds below are integrable. Then

$$0 \leq h_x(t) - J^\pi(t, x) \leq \int_t^T \max_{z \in \mathcal{X}} \frac{[b_1(s) + \chi_z(s)]^2}{2m_z^-(s)} ds. \quad (33)$$

The right side uses the computed v , the implemented quotes, and the supplied bounds; it contains no true continuation value or true state-occupation distribution.

Proof. At a common continuation vector v , the two Hamiltonians differ by at most $c(s)$, because each $v_y - v_x$ is bounded by $\text{span}(v)$. The vectors $v \pm D\mathbf{1}$ have the required terminal inequalities. Their true HJB residuals have opposite signs since $D' = -(\rho + c)$. The same comparison argument as in Lemma 6.3 brackets the true value by those vectors.

Expand the full discrepancy as

$$\mathcal{H}_x(a; h) - \hat{\mathcal{H}}_x(a; v) = (r_x - \hat{r}_x) + [(L^a - \hat{L}^a)v]_x + [L^a(h - v)]_x.$$

Its j -th action derivative is bounded by b_j . For $j = 2$, the symmetric-Hessian perturbation inequality gives true strong concavity with modulus m_x^- .

Choose the closest $n \in N_{\mathcal{A}_x}(\tilde{a})$ to the estimated gradient. For a true maximizing action a^* , strong concavity gives, with $d = a^* - \tilde{a}$,

$$\mathcal{H}_x(a^*; h) - \mathcal{H}_x(\tilde{a}; h) \leq (b_1 + \chi_x)\|d\| - \frac{m_x^-}{2}\|d\|^2 \leq \frac{(b_1 + \chi_x)^2}{2m_x^-}.$$

Here $n^\top d \leq 0$ removes the normal component. Apply Classical result 3.2 and then bound its occupation average by the maximum over states to obtain (33). $\square$

A piecewise C^1 , continuous interpolant is enough if all derivative inequalities hold almost everywhere. An approximate maximization in the definition of ρ must be enclosed by a valid upper and lower bound. Sampling the HJB residual or Hessian on a grid, without control between nodes, does not establish the supremum required here. For random model estimates the theorem holds on the event where all primitive bounds hold simultaneously; adaptive model selection requires bounds valid for the selected model. A fitted Hessian alone is not a lower bound for the true curvature.

Corollary 7.2 (A computed discounted certificate). *In the stationary discounted setting, let v be a computed state vector and use stationary primitive derivative bounds as above. Put*

$$\begin{aligned} \rho_\beta &\geq \max_x \left| \beta v_x - \max_a \hat{\mathcal{H}}_x(a; v) \right|, \\ D_\beta &= \frac{\rho_\beta + \epsilon_{r,0} + \text{span}(v)\epsilon_{k,0}}{\beta}, \\ b_{j,\beta} &= \epsilon_{r,j} + \text{span}(v)\epsilon_{k,j} + 2D_\beta \bar{K}_j, \quad j = 1, 2. \end{aligned}$$

Then $\|h - v\|_\infty \leq D_\beta$. If $-\nabla_a^2 \hat{\mathcal{H}}_x(a; v) \succeq \hat{m}_x I$ uniformly on the common convex action set and $m_x^- := \hat{m}_x - b_{2,\beta} > 0$, any stationary implemented quote with normal-cone residual χ_x satisfies

$$0 \leq h_x - J_\beta^\pi(x) \leq \frac{1}{\beta} \max_z \frac{(b_{1,\beta} + \chi_z)^2}{2m_z^-}.$$

Proof. The vectors $v \pm D_\beta \mathbf{1}$ are respectively a discounted supersolution and subsolution for the true Hamiltonian. The comparison in Theorem 6.6 gives the value enclosure. The derivative and normal-cone argument of Theorem 7.1 then bounds each true local gap by the displayed quadratic expression. Integrate (30) and bound its occupation average by the maximum over states. $\square$

7.1 A discrete-message certificate from value intervals

For a finite message menu, it is possible to retain state dependence without estimating action derivatives or assuming curvature. Consider the stationary discounted version of Proposition 3.9. Let $\theta \in \Theta$ index the possible external event laws, with generator $L^{0,\theta}$ and reward rate $r^{0,\theta}$. The decision rate ν , admissible message maps R_b , and fees $c_x(b)$ are known and common to the models. Put

$$H_x^{0,\theta}(v) = r_x^{0,\theta} + (L^{0,\theta}v)_x.$$

Assume $\beta > 0$, a common bound R_∞ on the absolute total reward rate, and a finite $\Lambda > 0$ bounding the external exit rate plus ν . All statements below are conditional on $\theta \in \Theta$. Define two operators by

$$(T^\pm v)_x = \frac{\Lambda v_x + \text{ext}_{\theta \in \Theta}^\pm H_x^{0,\theta}(v) + \nu \max_b \{v_{R_b(x)} - v_x - c_x(b)\}}{\beta + \Lambda}, \quad \text{ext}^- = \inf, \quad \text{ext}^+ = \sup. \quad (34)$$

Uniformization and uncertain transition laws are standard dynamic programming tools; robust dynamic programming with rectangular ambiguity is developed by [Iyengar \(2005\)](#). Here the extremum at each state is used to enclose every model-specific optimal value. The bounds remain valid if the rowwise uncertainty set enlarges a correlated parameter family. They do not require a single parameter vector to attain every rowwise extremum.

Theorem 7.3 (Value intervals and occupation-sensitive message loss). *Start with $l^0 = -R_\infty \mathbf{1}/\beta$ and $u^0 = R_\infty \mathbf{1}/\beta$, and iterate $l^{k+1} = T^- l^k$, $u^{k+1} = T^+ u^k$. Then*

$$l^k \leq h^\theta \leq u^k \quad (\theta \in \Theta, k \geq 0).$$

Rounding each lower update downward and each upper update upward preserves this conclusion. For any resulting enclosure $[l, u]$ and stationary message policy π , set

$$S_{yz}(l, u) = \begin{cases} u_y - l_z, & y \neq z, \\ 0, & y = z, \end{cases} \quad G_x = \nu \max_b \{S_{R_b(x), R_{\pi(x)}(x)}(l, u) - c_x(b) + c_x(\pi(x))\}.$$

Then $G_x \geq 0$. Write $L^{\pi,\theta}$ for the complete generator under that policy. Every bounded vector z satisfying

$$\beta z_x \geq G_x + \sup_{\theta \in \Theta} (L^{\pi,\theta} z)_x \quad (35)$$

certifies

$$0 \leq h_x^\theta - J_\beta^{\pi,\theta}(x) \leq z_x \quad (\theta \in \Theta).$$

Alternatively, start $z^0 = \|G\|_\infty \mathbf{1}/\beta$ and iterate

$$z_x^{k+1} = \frac{G_x + \Lambda z_x^k + \sup_{\theta \in \Theta} (L^{\pi,\theta} z^k)_x}{\beta + \Lambda}. \quad (36)$$

Every iterate is a valid upper bound for the policy loss, including updates rounded upward. If $G \equiv 0$, the policy is optimal simultaneously for every model in the declared family.

Proof. For a fixed model and message, uniformization writes the numerator of the Bellman map as the reward rate plus a nonnegative linear combination of continuation values whose coefficients sum to Λ . Thus $T^\pm$ are monotone contractions in sup norm with modulus $\Lambda/(\beta + \Lambda)$. The model-specific map T^θ obeys $T^- \leq T^\theta \leq T^+$, and its fixed point is h^θ . The reward bound brackets this fixed point at iteration zero. Monotonicity proves the value enclosure by induction, also with outward rounding.

At h^θ , the external Hamiltonian is common to all messages at the current state. It cancels from the action gap:

$$\Delta_x^\theta = \nu \max_b \{h_{R_b(x)}^\theta - h_{R_{\pi(x)}(x)}^\theta - c_x(b) + c_x(\pi(x))\} \leq G_x.$$

When the two destinations agree, their continuation difference is exactly zero; replacing it by $u_y - l_y$ would introduce an unnecessary error. The deployed message itself contributes zero, which gives $G_x \geq 0$.

The discounted performance identity gives

$$h^\theta - J_\beta^{\pi, \theta} = (\beta I - L^{\pi, \theta})^{-1} \Delta^\theta \leq (\beta I - L^{\pi, \theta})^{-1} G.$$

The resolvent is positive because it is the integral of the discounted transition semigroup. Applying it to (35) proves the supersolution claim. The fixed-policy map with reward G is monotone, is bounded above by the map in (36), and has fixed point at most $\|G\|_\infty/\beta$. Induction proves the finite-iterate claim. Upward rounding need not leave an iterate an exact supersolution; the comparison with every fixed-policy value nevertheless remains valid. Finally $G = 0$ forces every true action gap to vanish. $\square$

This calculation retains where a wrong message can occur and how often the deployed policy visits those states. The uniform bound $\|G\|_\infty/\beta$ discards that information. Neither bound requires the true policy occupation measure as input. For an event with known destination y , uncertain rate $\lambda \in [\lambda^-, \lambda^+]$, and uncertain net mark $g \in [g^-, g^+]$, its external Hamiltonian contribution is $\lambda(g + v_y - v_x)$. Its exact extrema over this rectangle occur among the four corners. Using this expression preserves the fact that the same event rate multiplies both the mark and the continuation difference.

8 Improving a baseline quoting policy under uncertainty

The preceding certificates bound loss relative to a model-specific optimum. A different deployment question compares a candidate with the policy already running. Baseline-relative robust policy improvement is established by Ghavamzadeh et al. (2016). Here its continuous-time specialization uses the tagged queue state, a supplied enclosure of the baseline value, and an explicit lower bound on the improvement. It does not require the unknown optimal value or an online exploration claim.

8.1 An advantage measured against one common law

Fix a finite state and action space, discount $\beta > 0$, and a family $\theta \in \Theta$ of bounded reward rates $r^{a, \theta}$ and generators $L^{a, \theta}$. The state includes every outstanding reservation and any pending cancellation. Let $b(x)$ be a stationary baseline and

$$V^{b, \theta} = (\beta I - L^{b, \theta})^{-1} r^{b, \theta}.$$

Suppose vectors l, u satisfy $l \leq V^{b, \theta} \leq u$ for every θ . For a row difference $D_{xy}^{a, \theta} = L_{xy}^{a, \theta} - L_{xy}^{b, \theta}$, define

$$\underline{A}_x(a) = \inf_{\theta \in \Theta} \left\{ r_x^{a, \theta} - r_x^{b, \theta} + \sum_{y: D_{xy}^{a, \theta} \geq 0} D_{xy}^{a, \theta} l_y + \sum_{y: D_{xy}^{a, \theta} < 0} D_{xy}^{a, \theta} u_y \right\}. \quad (37)$$

This is the exact minimum over the supplied value rectangle and the parameter family; the rectangle can be an enlargement of the attainable baseline values. The baseline action has $\underline{A}_x(b(x)) = 0$ exactly. The same θ must be used in both terms of each reward and generator difference. Separately minimizing the candidate and maximizing the baseline can destroy cancellations and make the bound needlessly pessimistic.

Theorem 8.1 (A certified baseline policy improvement). *Choose a stationary policy π with $g_x := \underline{A}_x(\pi(x)) \geq 0$ for every state, retaining $b(x)$ when no strictly positive alternative is certified. Then, for every $\theta \in \Theta$,*

$$V^{\pi, \theta} - V^{b, \theta} \geq (\beta I - L^{\pi, \theta})^{-1} g \geq 0. \quad (38)$$

Any vector z satisfying

$$\beta z_x \leq g_x + \inf_{\theta \in \Theta} (L^{\pi, \theta} z)_x \quad (39)$$

is a simultaneous lower bound on this value improvement. With a common exit-rate bound Λ , start $z^0 = 0$ and iterate

$$z_x^{k+1} = \frac{g_x + \Lambda z_x^k + \inf_{\theta \in \Theta} (L^{\pi, \theta} z^k)_x}{\beta + \Lambda}. \quad (40)$$

Each iterate is a lower bound. Downward rounding preserves the bound; replacing a negative rounded coordinate by zero also preserves it.

Proof. Under one fixed law, the baseline advantage is $A^{\pi, b, \theta} = r^{\pi, \theta} - r^{b, \theta} + (L^{\pi, \theta} - L^{b, \theta})V^{b, \theta}$. Subtract the two policy equations to obtain the exact identity

$$(\beta I - L^{\pi, \theta})(V^{\pi, \theta} - V^{b, \theta}) = A^{\pi, b, \theta}.$$

Minimizing a linear form over $[l, u]$ proves $A^{\pi, b, \theta} \geq g$. The resolvent is the integral of a positive discounted transition semigroup, proving (38). Multiplying the subsolution inequality by this positive resolvent proves its assertion. Uniformization makes the iteration monotone, with contraction factor $\Lambda/(\beta + \Lambda)$. It lies below each model's fixed-policy map with reward g . Starting at zero and induction therefore bound its value, including downward rounding. Nonnegativity permits clipping a rounded lower bound at zero. Combining these facts proves the finite-iterate statement. $\square$

Remark 8.2 (Policy value, not realized pathwise profit). The comparison is between discounted expectations from every modeled state. It does not imply that every realized trading path improves, that withdrawal liquidates inventory, or that a nominally safe action remains safe after the law leaves Θ . Rectangular enlargement is allowed when computing a lower bound for a fixed parameter family. It need not represent a single law attaining every rowwise infimum. All actions must share valid physical state transitions, including executable orders during cancellation latency.

8.2 The message-clock specialization

In the replacement-clock model the external events are common to the messages at a given state. A message a sends x to $R_a(x)$ at known rate ν and costs $c_x(a)$. Write $y = R_a(x)$ and $z = R_{b(x)}(x)$. External rewards, external generators, and the diagonal clock term cancel. Consequently a computable lower advantage is

$$\underline{A}_x(a) = \nu \left[\begin{cases} l_y - u_z, & y \neq z, \\ 0, & y = z \end{cases} - c_x(a) + c_x(b(x)) \right]. \quad (41)$$

When destinations agree their value difference is exactly zero, regardless of the enclosure width. This cancellation is particularly useful when two message labels induce the same physical transition.

The bounds $[l, u]$ must enclose the *baseline* value, not the optimal value from Theorem 7.3. They are obtained by fixing the message $b(x)$ in that theorem’s lower and upper uniformized recursions. The same monotonicity proof applies. An independent alternative starts from a computed baseline value v . If

$$\sup_{x, \theta} |r_x^{b, \theta} + (L^{b, \theta} v)_x - \beta v_x| \leq \eta,$$

then $v - \eta \mathbf{1}/\beta \leq V^{b, \theta} \leq v + \eta \mathbf{1}/\beta$ by the resolvent norm bound. This accounts for both model discrepancy and a numerical policy-evaluation residual.

Corollary 8.3 (Randomization, repeated improvement, and tolerance). *Any statewise mixture supported on actions with nonnegative certified advantage also improves on b . A finite succession of such improvements, each with a fresh valid enclosure for its new baseline and the same family Θ , is monotone in value for every $\theta \in \Theta$. If numerical errors are bounded by $e_x(a)$, replace an approximate advantage by its certified lower endpoint before screening actions.*

Proof. Rewards and generators are affine in statewise action probabilities, so the advantage is the corresponding convex combination. Apply the theorem and then transitivity to successive baseline comparisons. Certified lower endpoints preserve the required nonnegativity. $\square$

Repeated improvement need not reach a globally optimal policy. A wide enclosure can block every nontrivial change; this is a meaningful output. Nor does choosing actions greedily with an estimated baseline value alone provide this guarantee. Near a switching boundary even a small value error can reverse a message comparison.

8.3 A queue-aware baseline experiment

The experiment reuses the 74 states and all admissible messages of Section 12.5. The nominal optimum is used to construct the baseline. That baseline deliberately discards tagged rank: at a back-of-queue state, it applies the action chosen at the otherwise identical front state. This reproduces a concrete information omission, not an arbitrarily poor constant policy. For each uncertainty multiplier the script computes 320 lower and upper baseline-value iterations with exact fractions and 12-decimal outward rounding. It evaluates all admissible messages and accepts positive certified changes and computes 320 lower-gain iterations.

Table 1: Synthetic queue experiment. Gains are discounted model objective units from the empty, zero-inventory state in regime zero. The lower bound covers the entire stated box; the last column is a nominal-model evaluation.

Uncertainty	Changed states	Certified gain	Nominal gain
0	13	0.00016417	0.00016417
1	13	0.00016143	0.00016417
4	13	0.00015324	0.00016417
16	10	0.00012083	0.00016417
64	7	0.00005378	0.00011996

The script `verification/certified_improvement.py` independently evaluates both policies by solving their linear systems at the center, at aligned lower and upper corners, and at 20 seeded,

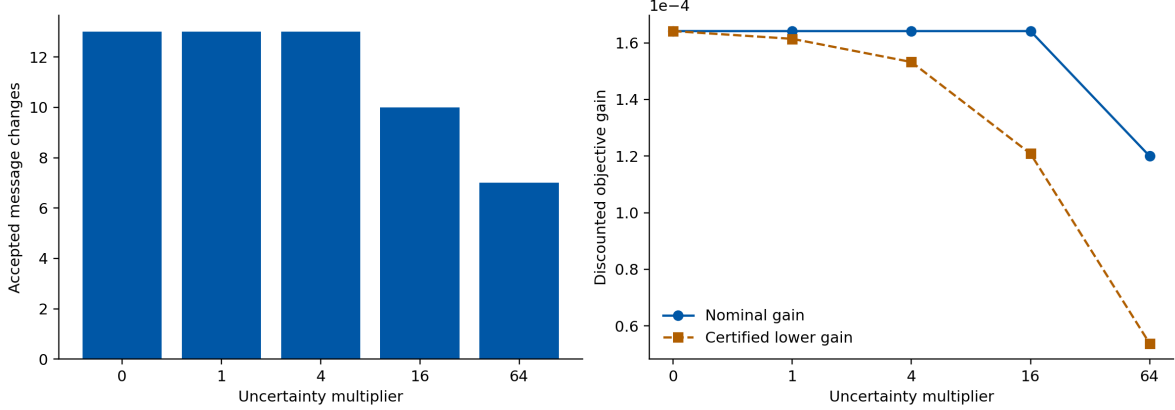


Figure 3: A baseline policy can be improved selectively as queue-law uncertainty changes. Certified gains use rational outward enclosures; nominal gains use independent linear policy evaluation. The plotted experiment is synthetic and excludes empirical profitability claims.

independently varied event-law draws. It verifies the exact performance-difference identity and the lower-gain comparison in every state. These sampled comparisons are diagnostics; coverage of the whole box comes from the proved enclosure and rational arithmetic, not from sampling. A zero-uncertainty comparison also tests the contraction remainder rather than silently treating a finite iteration as an exact fixed point.

To turn this into market evidence, log receipt, decision, send, acknowledgement, fill and cancellation times, and retain an order’s tagged priority or its identifiable bounds. Estimate state- and action-conditional event laws on training days, freeze the baseline and screened policy, and report acceptance frequency together with held-out inventory, markout, fees and net P&L. Off-policy logs without coverage cannot identify counterfactual fills; the screening theorem does not repair that identification problem.

9 Hybrid actions and the cost of a small decision margin

Let the action set be a finite disjoint union of labelled convex compact branches $A_{x,j}$, with at least two branches when the following gap is used. A branch can fix which sides are active; singleton branches represent individual ticks. Write

$$V_j(s, x) = \max_{a \in A_{x,j}} \mathcal{H}_x(a; h(s)).$$

Let $\Gamma(s, x)$ be the difference between the largest and the second largest of these values, with $\Gamma = 0$ at a tie. With one branch put $\Gamma = \infty$.

Theorem 9.1 (Hybrid margin certificate). *Suppose the full estimated and true Hamiltonians differ uniformly by at most ζ , and their within-branch gradients by at most b . Suppose each true branch is m -strongly concave, $m > 0$, and the estimated policy maximizes over all branches. For its true occupation law define*

$$\mathcal{O}_{\hat{\pi}}(z) = \mathbb{E}_{t,x}^{\hat{\pi}, \text{true}} \int_t^T \mathbf{1}_{\{\Gamma(s, X_s) \leq z\}} ds.$$

Then

$$h_x(t) - J^{\hat{\pi}}(t, x) \leq \frac{(T-t)b^2}{2m} + 2\zeta \mathcal{O}_{\hat{\pi}}(2\zeta). \quad (42)$$

In particular, if $\mathcal{O}_{\hat{\pi}}(z) \leq C_{\text{occ}} z^\alpha$ for $0 < z \leq z_0$, where $\alpha > 0$ and $2\zeta \leq z_0$, the switching contribution is at most $2^{\alpha+1} C_{\text{occ}} \zeta^{\alpha+1}$. For singleton branches the first term can be set to zero.

Proof. Taking a maximum within a branch changes its value by at most ζ . If $\Gamma > 2\zeta$, the estimated maximizing branch must be the unique true maximizing branch. Lemma 6.1 then bounds the within-branch loss by $b^2/(2m)$. On the complementary event, Proposition 6.5 bounds the entire local gap by 2ζ . The sum of the two bounds with the second multiplied by $\mathbf{1}_{\{\Gamma \leq 2\zeta\}}$ is therefore valid everywhere. Integrate the performance identity. On a singleton branch there is no within-branch displacement. $\square$

If $b, \zeta = O(\varepsilon)$, the rate is $O(\varepsilon^2 + \varepsilon^{\alpha+1})$. This condition is on the states and times visited by the deployed policy in the true model. A margin distribution under a logging policy cannot replace it without a separate occupation comparison. Uniform asymptotic rates require uniform occupation constants over the family of fitted policies. A positive lower decision gap gives eventual branch preservation directly. The argument is the dynamic counterpart of margin reasoning for plug-in classifiers (Audibert and Tsybakov, 2007).

Example 9.2 (Sharp switching exponent). Consider two singleton actions at a random observed covariate $\Gamma \in [0, g_0]$, with true rewards $0, \Gamma$ and estimated rewards $\zeta, \Gamma - \zeta$. Both value errors are exactly ζ . Suppose $\mathbb{P}(\Gamma \leq z) = (z/g_0)^\alpha$, $0 \leq z \leq g_0$, with $\alpha > 0$. For $2\zeta \leq g_0$, the loss is

$$\mathbb{E}[\Gamma \mathbf{1}_{\{\Gamma < 2\zeta\}}] = \frac{\alpha}{\alpha + 1} \frac{(2\zeta)^{\alpha+1}}{g_0^\alpha}.$$

Thus the margin exponent cannot be improved over this class of one-decision problems. The within-branch gradient error is zero. This covariate example establishes sharpness of the boundary mechanism; it does not assert an irreducible finite-state occupation law with that density.

10 Why a fill certificate requires action coverage

Derivative error budgets concern actions that a quoting policy might choose. A large data set at one offset need not identify them. This is an instance of the coverage problem in offline decision making (Jin et al., 2021); the following construction gives its cost explicitly in a quoting cell.

Proposition 10.1 (Indistinguishable logs, different optimal quotes). *Consider the static gross-revenue criterion $F_\kappa(\delta) = \delta \Lambda_\kappa(\delta)$ on $A = [0, 3/5]$, with*

$$\Lambda_\kappa(\delta) = \lambda_0 e^{-\kappa(\delta-2/5)}, \quad \kappa \in \{2, 3\}, \quad \lambda_0 > 0.$$

Suppose the calibration data consist only of a Poisson fill log generated at the fixed offset $2/5$, with identical deterministic marks in the two models. Every observation duration and number of independent repetitions has the same data law in the two models. Nevertheless their unique optimal offsets are $\delta_2^ = 1/2$ and $\delta_3^* = 1/3$. For every data-dependent, possibly randomized quote $\hat{\delta}$,*

$$\max_{\kappa \in \{2, 3\}} \mathbb{E}_\kappa[F_\kappa(\delta_\kappa^*) - F_\kappa(\hat{\delta})] \geq \frac{m_*}{8} \left(\frac{1}{2} - \frac{1}{3} \right)^2 > 0, \quad (43)$$

where

$$m_* = \lambda_0 \min_{\kappa \in \{2, 3\}} e^{-\kappa/5} \kappa (2 - 3\kappa/5) > 0.$$

Proof. Both logged intensities equal λ_0 , so the complete observed point-process laws agree. The derivative of F_κ has its unique zero at $1/\kappa$, inside A . Moreover $-F_\kappa''(\delta) = \Lambda_\kappa(\delta)\kappa(2 - \kappa\delta)$ is positive and decreasing on A , and hence bounded below by m_* . True regret is at least $m_*(\hat{\delta} - \delta_\kappa^*)^2/2$. The quote has the same distribution under both data laws. For any real z ,

$$(z - \delta_2^*)^2 + (z - \delta_3^*)^2 \geq (\delta_2^* - \delta_3^*)^2/2.$$

Sum the expected loss bounds and divide by two to bound their maximum, obtaining (43). $\square$

The lower bound is about the declared fixed logging design, not about every exploration scheme. Varying the offset can distinguish these parametric laws. Likewise, observing a noiseless response to one parent execution is a different inverse problem from inferring counterfactual fills at untried quotes. Cost-relevant identification depends on the experiment as well as the decision.

10.1 A positive certificate from two logging quotes

Consider the static cell

$$\Lambda_\kappa(\delta) = \lambda_0 e^{-\kappa(\delta - \delta_0)}, \quad F_\kappa(\delta) = \delta \Lambda_\kappa(\delta), \quad A = [0, D].$$

Both $\lambda_0, \kappa > 0$ are unknown. Assume $1/\kappa < D \leq 2/\kappa$, and choose feasible logging offsets $0 \leq \delta_0 < \delta_1 = \delta_0 + \Delta \leq D$. Each is held for a deterministic duration τ , in a stationary Poisson experiment with independent counts

$$N_0 \sim \text{Poisson}(m_0), \quad N_1 \sim \text{Poisson}(m_1), \quad m_0 = \tau \lambda_0, \quad m_1 = \tau \lambda_0 e^{-\kappa \Delta}.$$

The total logging duration is 2τ . When both counts are positive, set $\hat{\kappa} = \Delta^{-1} \log(N_0/N_1)$. If also $\hat{\kappa} > 0$, deploy $\hat{\delta} = \text{clip}_{[0, D]}(1/\hat{\kappa})$; otherwise set $\hat{\delta} = 0$. The scale estimate N_0/τ and the slope estimate jointly identify the cell asymptotically, although only the slope enters its optimizer.

This experiment assumes a stationary Poisson fill law at each quote. For a tagged queue, Proposition 3.4 shows why that assumption does not follow from a fixed price alone. Applying this statistical certificate to a queue-based strategy requires a justified conditional-intensity model and the associated queue observations.

Proposition 10.2 (Two-quote revenue certificate). *Fix $\epsilon \in (0, 1)$, and set*

$$t = \sqrt{\frac{3 \log(4/\epsilon)}{m_1}}, \quad M = 2\kappa \lambda_0 e^{\kappa \delta_0}.$$

If $t \leq \min\{1/2, \kappa \Delta/8\}$, then, with probability at least $1 - \epsilon$,

$$\begin{aligned} F_\kappa(1/\kappa) - F_\kappa(\hat{\delta}) &\leq \frac{8Mt^2}{\Delta^2 \kappa^2 (\kappa - 4t/\Delta)^2} \\ &\leq \frac{64e^{\kappa(\delta_0 + \Delta)}}{\Delta^2 \kappa^3} \frac{3 \log(4/\epsilon)}{\tau}. \end{aligned} \tag{44}$$

This is a bound on the deployed revenue loss per unit time, excluding revenue forgone during the logging experiment.

Proof. The Poisson Chernoff inequalities give, for $0 < t < 1$,

$$\mathbb{P}(|N/m - 1| > t) \leq 2e^{-mt^2/3}.$$

Since $m_0 \geq m_1$, the event $E = \{|N_i/m_i - 1| \leq t, i = 0, 1\}$ has probability at least $1 - 4e^{-m_1 t^2/3} = 1 - \epsilon$. Both counts are positive on this event. Using $|\log(1 + u)| \leq 2|u|$ for $|u| \leq 1/2$ gives

$$|\hat{\kappa} - \kappa| \leq 4t/\Delta \leq \kappa/2, \quad |1/\hat{\kappa} - 1/\kappa| \leq \frac{4t/\Delta}{\kappa(\kappa - 4t/\Delta)}.$$

Projection onto $[0, D]$, which contains the true optimizer $1/\kappa$, cannot increase this distance. On this interval

$$-F''_{\kappa}(\delta) = \kappa\Lambda_{\kappa}(\delta)(2 - \kappa\delta) \quad \text{lies in } [0, M].$$

Indeed it is nonnegative and decreasing there, with value M at zero. Taylor's theorem around the interior optimum yields $F_{\kappa}(1/\kappa) - F_{\kappa}(\hat{\delta}) \leq M(\hat{\delta} - 1/\kappa)^2/2$, proving the first bound. Use $\kappa - 4t/\Delta \geq \kappa/2$ and $t^2 = 3\log(4/\epsilon)e^{\kappa\Delta}/(\tau\lambda_0)$ to obtain the second. The scale cancels from this upper bound. $\square$

The concentration condition still depends on the unknown true parameters. A design with a guaranteed sample size needs prior lower bounds on the far-quote intensity and slope, or a separately valid confidence construction. For example, known $\kappa \in [\kappa_-, \kappa_+]$ permits the common concave quote range above if $1/\kappa_- < D \leq 2/\kappa_+$.

Corollary 10.3 (Expected loss and feasible spacing). *For fixed parameters and fixed feasible $\Delta > 0$, as $\tau \rightarrow \infty$,*

$$\mathbb{E}[F_{\kappa}(1/\kappa) - F_{\kappa}(\hat{\delta})] = \frac{e^{\kappa\delta_0-1}(1 + e^{\kappa\Delta})}{2\Delta^2\kappa^3\tau} + o(\tau^{-1}). \quad (45)$$

For fixed δ_0, D, κ under the stated concave-range constraint, this leading constant is minimized over the feasible spacing range by $\Delta = D - \delta_0$.

Proof. On any fixed small relative-count neighborhood of the true means, the log ratio, its reciprocal, and the revenue loss are smooth; the clipping and fallback are inactive. Independence and the Poisson variance formula give the leading slope variance

$$\frac{1 + e^{\kappa\Delta}}{\tau\lambda_0\Delta^2}.$$

Taylor expansion of the reciprocal contributes κ^{-4} to the squared offset error, and $-F''_{\kappa}(1/\kappa) = \kappa\lambda_0e^{\kappa\delta_0-1}$. The third absolute moments of the relative-count errors are $O(\tau^{-3/2})$. Chernoff bounds make the complementary event exponentially unlikely, while the clipped revenue loss is bounded. These estimates justify the expectation expansion and its $o(\tau^{-1})$ remainder, giving (45).

With $x = \kappa\Delta$, the spacing-dependent factor is $(1 + e^x)/x^2$. Its derivative has the sign of $e^x(x - 2) - 2$, whose unique positive zero is $x_* \approx 2.2177$, equivalently $x_* = 2(1 + e^{-x_*})$. Feasibility implies $x \leq \kappa(D - \delta_0) \leq 2 < x_*$, so this factor decreases throughout the feasible interval. The unconstrained stationary point is outside the model's permitted quote range. $\square$

With $\lambda_0 = 1$, $D = 3/5$, and $\delta_0 = 2/5$, logging also at $3/5$ distinguishes the two laws in Proposition 10.1. For $\kappa = 2, 3$, respectively, the leading expected-loss constants in (45) are approximately 3.1877 and 1.5958, in units of revenue loss times logging duration τ . Two quotes therefore remove the fixed-design identification floor within this parametric cell. This calculation assumes a stationary exponential family and does not identify an unrestricted fill law from two offsets.

10.2 Information and sampling lower bounds inside a queue

Action coverage is not the only obstacle to a small certificate. An observation can omit the queue variable that determines the preferred action. Even with that variable observed, a finite sample can leave the sign of a replacement advantage unresolved.

Proposition 10.4 (The loss of omitting tagged rank). *Consider the queue cell of Proposition 3.4, with $\mu = 1$, no cancellations, $\beta + \xi = 1$, and execution reward $R = 1/2$. The tagged unit has either zero or two lots ahead. In both states the displayed depth is four lots including that unit, and the quote price and all other supplied observations agree. The available decisions are to keep until fill or invalidation, or to take a known alternative of value $K = 33/200$. For a policy restricted to this common snapshot,*

$$\inf_{\pi} \max_{n \in \{0,2\}} \{\text{optimal value at rank } n - \text{value of } \pi\} = \frac{697}{15000}.$$

The infimum includes randomized decisions and is attained by keeping with probability $34/75$.

Proof. Keeping is worth $1/4$ at rank zero and $1/16$ at rank two. Its advantage in the first state is $a = 17/200$; the alternative's advantage in the second is $b = 41/400$. A snapshot policy must use the same keeping probability p in both states. Its worst loss is $\max\{(1-p)a, pb\}$, minimized when the two terms agree. Thus $p = a/(a+b) = 34/75$, and the loss is $ab/(a+b) = 697/15000$. $\square$

This positive lower bound persists with exact knowledge of every rate and reward. It concerns the stated snapshot information class: an order-message history that recovers tagged rank can remove it. The alternative can be the one-time replacement described after (9); subsequent switching is excluded.

Theorem 10.5 (A sharp calibration rate at a replacement boundary). *Suppose rank $n \geq 0$ is observed, $R > 0$ and $\vartheta = \beta + \xi > 0$ are known, and ahead cancellations are absent. Keeping has value*

$$A_n(\mu) = R \left(\frac{\mu}{\mu + \vartheta} \right)^{n+1}.$$

The alternative has known value $K = A_n(\mu_0)$, where $0 < \mu_- < \mu_0 < \mu_+ < \infty$. An independent stationary calibration experiment observes market orders reaching this level for duration τ , giving $N \sim \text{Pois}(\tau\mu)$. Let $\mathcal{R}_\tau(\pi, \mu)$ denote the expected post-calibration value loss of a rule choosing between keeping and the alternative. Put

$$L = \max_{\mu \in [\mu_-, \mu_+]} A'_n(\mu), \quad a_* = \min_{\mu \in [\mu_-, \mu_+]} A'_n(\mu) > 0, \quad d_0 = \frac{\sqrt{\mu_-}}{8}.$$

For all sufficiently large τ ,

$$\frac{3a_*d_0}{8\sqrt{\tau}} \leq \inf_{\pi} \sup_{\mu \in [\mu_-, \mu_+]} \mathcal{R}_\tau(\pi, \mu) \leq L\sqrt{\frac{\mu_+}{\tau}}. \quad (46)$$

The upper bound is attained as a bound by the plug-in rule that keeps when $A_n(\hat{\mu}) \geq K$, with $\hat{\mu} = \text{clip}_{[\mu_-, \mu_+]}(N/\tau)$. For fixed $z \neq 0$, under $\mu_\tau = \mu_0 + z\sqrt{\mu_0/\tau}$, that rule also satisfies

$$\sqrt{\tau} \mathcal{R}_\tau(\pi, \mu_\tau) \longrightarrow A'_n(\mu_0)\sqrt{\mu_0} |z| \Phi(-|z|), \quad (47)$$

where Φ is the standard normal distribution function.

Proof. The derivative

$$A'_n(\mu) = \frac{R(n+1)\vartheta\mu^n}{(\mu + \vartheta)^{n+2}}$$

is positive and continuous on the compact parameter interval. A wrong plug-in action implies that K lies between $A_n(\mu)$ and $A_n(\hat{\mu})$, so its loss is at most $L|\mu - \hat{\mu}|$. Clipping cannot increase estimation error. Cauchy–Schwarz and the Poisson variance give $\mathbb{E}|\mu - \hat{\mu}| \leq \sqrt{\mu/\tau}$, proving the upper bound.

For the lower bound choose $\mu^\pm = \mu_0 \pm d_0/\sqrt{\tau}$, which lie in the parameter interval for sufficiently large τ . Their Poisson laws $P_\pm$ satisfy

$$\text{KL}(P_- \| P_+) = \tau \{ \mu^- \log(\mu^- / \mu^+) - \mu^- + \mu^+ \} \leq \frac{\tau(\mu^- - \mu^+)^2}{\mu^+} \leq \frac{1}{16}.$$

Pinsker's inequality yields $\text{TV}(P_-, P_+) \leq \sqrt{\text{KL}(P_- \| P_+)/2} \leq 1/\sqrt{32} < 1/4$. Keeping is wrong under μ^- and replacing is wrong under μ^+ ; any randomized test has sum of these two error probabilities at least $1 - \text{TV}(P_-, P_+) > 3/4$. Each wrong-action gap is at least $a_* d_0 / \sqrt{\tau}$. The larger of the two risks is at least half their sum, proving the lower bound. Conditional arrival times given the count have a parameter-free law, so observing the full Poisson history does not defeat this argument.

Under the displayed local sequence, a Poisson central limit theorem gives wrong-action probability tending to $\Phi(-|z|)$. Taylor expansion of $|A_n(\mu_\tau) - A_n(\mu_0)|$, multiplied by $\sqrt{\tau}$, gives the remaining factor in (47). $\square$

The $\tau^{-1/2}$ minimax rate is a discrete replacement effect. It differs from the quadratic post-calibration loss on a uniformly curved quote branch. The experiment observes market orders rather than tagged fills, and the result concerns one calibrated decision; it is not a statement about cumulative online learning regret.

11 Signal-value ceilings for bounded and unbounded rewards

11.1 A bounded reward

Classical result 11.1 (Value of a signal). *Let ϑ be a state and Y a signal on standard Borel spaces, and let $r(a, \vartheta)$ be a measurable one-decision reward with global range at most Δ . Assume finite action sets, or measurable approximate maximizing selectors. Then the value of observing Y obeys*

$$0 \leq \mathbb{E}_Y \sup_a \mathbb{E}[r(a, \vartheta) \mid Y] - \sup_a \mathbb{E}r(a, \vartheta) \leq \Delta \mathbb{E}_Y \text{TV}(P_{\vartheta|Y}, P_\vartheta) \leq \Delta \sqrt{I(\vartheta; Y)/2}.$$

Information is measured in nats.

Proof. Write $f_y(a) = \mathbb{E}[r(a, \vartheta) \mid Y = y]$ and $f(a) = \mathbb{E}r(a, \vartheta)$. The elementary inequality

$$\sup_a f_y(a) - \sup_a f(a) \leq \sup_a (f_y(a) - f(a))$$

and the bounded-range total-variation inequality give the first upper bound after averaging. Pinsker's inequality and Jensen's inequality give the second. Ignoring the signal proves nonnegativity; approximate maximizers handle nonattainment. $\square$

This is the standard total-variation and information argument (Cover and Thomas, 2006). The exponential-jump model in Section 5 has unbounded realized losses, so it requires a different hypothesis. A concave upper bound in information does not imply that the actual value of successive signals has diminishing returns.

11.2 A centered exponential-moment condition

Let P be the law of ϑ and $Q_y = P_{\vartheta|Y=y}$. The action set is common to all signal values. Assume jointly measurable rewards, measurable approximate maximizing selectors, and integrability under P and Q_y for almost every y . Suppose $V_0 = \sup_a \mathbb{E}_P r(a, \vartheta)$ is finite and a common convex function $\psi : [0, s_0) \rightarrow [0, \infty)$, with $\psi(0) = 0$, obeys

$$\log \mathbb{E}_P \exp\{s[r(a, \vartheta) - \mathbb{E}_P r(a, \vartheta)]\} \leq \psi(s), \quad 0 \leq s < s_0, \quad (48)$$

uniformly over actions. Conditional expectations and the uniform bound are understood on common sets of full measure. Define

$$\psi^\dagger(t) = \inf_{0 < s < s_0} \frac{t + \psi(s)}{s}, \quad t \geq 0.$$

Theorem 11.2 (Signal value under a Cramér-type bound). *Under these assumptions, if $I(\vartheta; Y) < \infty$, then*

$$\begin{aligned} 0 \leq \mathbb{E}_Y \sup_a \mathbb{E}_{Q_Y} r(a, \vartheta) - V_0 &\leq \mathbb{E}_Y \psi^\dagger(\text{KL}(Q_Y \| P)) \\ &\leq \psi^\dagger(I(\vartheta; Y)). \end{aligned} \quad (49)$$

The conditional supremum is integrable. If the signal value is well-defined and information is infinite, the bound remains valid with the usual infinite upper bound.

Proof. For $Q \ll P$, the entropy variational inequality gives

$$s \mathbb{E}_{Q_Y} g \leq \text{KL}(Q \| P) + \log \mathbb{E}_P e^{s g}.$$

Use $g = r(a, \cdot) - \mathbb{E}_P r(a, \cdot)$, and optimize over s . The result is uniform in a , hence

$$\sup_a \mathbb{E}_{Q_Y} r(a, \vartheta) - V_0 \leq \sup_a [\mathbb{E}_{Q_Y} r(a, \vartheta) - \mathbb{E}_P r(a, \vartheta)] \leq \psi^\dagger(\text{KL}(Q_Y \| P)).$$

For a fixed s , the final quantity is bounded above by an affine function of the relative entropy and is therefore integrable when information is finite. Any fixed action supplies an integrable lower bound on the conditional supremum, proving its integrability. The function $\psi^\dagger$ is concave and nondecreasing as an infimum of affine functions with positive slopes. Averaging and Jensen's inequality prove both upper bounds. Ignoring the signal, or taking unconditional approximate maximizers, proves nonnegativity. $\square$

This is an application of the classical entropy variational principle. Related information bounds for adaptive selection are developed by [Russo and Zou \(2016\)](#); the role here is to express the reward-specific moment condition and the units of signal value.

Example 11.3 (Bounded and finite-radius moment bounds). A reward of range Δ has $\psi(s) = \Delta^2 s^2 / 8$, so $\psi^\dagger(t) = \Delta \sqrt{t/2}$, recovering Classical result [11.1](#). If instead $\sigma, b > 0$ and $\psi(s) = \sigma^2 s^2 / 2$ is valid only on $0 \leq s < 1/b$, then

$$\psi^\dagger(t) = \begin{cases} \sqrt{2\sigma^2 t}, & t \leq \sigma^2 / (2b^2), \\ bt + \sigma^2 / (2b), & t > \sigma^2 / (2b^2). \end{cases}$$

The second branch is the limit as $s \uparrow 1/b$; no exponential moment at the boundary is needed.

Example 11.4 (Unbounded exponential mark losses). At an informed opportunity in Definition 5.1, write

$$r(\delta, J) = (\delta - J)\mathbf{1}_{\{J > \delta\}}, \quad J \sim \text{Exp}(k), \quad \delta \geq 0.$$

This is the markout per opportunity, with zero payoff when there is no fill. Put $p = e^{-k\delta}$. Direct integration gives

$$\mathbb{E}r = -p/k, \quad \mathbb{E}e^{sr} = 1 - \frac{ps}{k+s}, \quad s \geq 0.$$

Consequently

$$\log \mathbb{E}e^{s(r-\mathbb{E}r)} = px + \log \left(1 - \frac{px}{1+x} \right), \quad x = s/k.$$

For fixed $x \geq 0$, this expression is increasing in $p \in [0, 1]$: its derivative is $x - x/[1 + (1-p)x] \geq 0$. Its maximum at $p = 1$ is $x - \log(1+x) \leq x^2/2$, the last inequality following by integration of $x/(1+x) \leq x$. Thus the uniform envelope $\psi(s) = s/k - \log(1+s/k)$ is valid. Its quadratic upper bound $s^2/(2k^2)$ gives $\sqrt{2I(J; Y)}/k$; taking $s \rightarrow \infty$ in its defining infimum also gives $\psi^\dagger(t) \leq 1/k$. The signal value is therefore at most

$$\frac{\min\{\sqrt{2I(J; Y)}, 1\}}{k},$$

in price units per informed opportunity. The bound is not a per-realized-fill statement: conditioning on a fill changes the law and the denominator. A deterministic action-dependent reward can be added without changing the centered moment bound, provided the theorem's finiteness conditions remain valid. This permits a nontrivial trade-off with spread revenue; with unrestricted withdrawal and this nonpositive markout alone, both optimal values are simply zero. Multiplication by a fixed exogenous opportunity rate gives a per-time bound only when each opportunity is the same one-decision problem.

12 Synthetic verification

12.1 A marked inventory model

The numerical supplement solves an inventory model with $q \in \{-2, -1, 0, 1, 2\}$, horizon $T = 1$, active offsets in $[0.15, 1]$, running penalty $0.02q^2$, and terminal penalty $0.05q^2$. A side is disabled when its fill would exceed an inventory limit. The total price compensator is zero, and each fill has expected co-jump loss ℓ . Active intensities are $\lambda_s e^{-\kappa\delta_s}$.

Parameter	True model	Estimated model
Ask intensity scale λ_a	2.20	2.35
Bid intensity scale λ_b	2.00	1.85
Offset decay κ	1.20	1.15
Co-jump loss ℓ	0.040	0.055

The optimal and estimated HJB equations are solved backward. The estimated policy is then evaluated under the true model by a separate backward linear equation. A forward state-probability equation computes the integrated true Hamiltonian gap and the gradient certificate. Starting at $q = 0$, these calculations give:

Quantity	Model units
Direct policy regret	0.0012786978
Integrated Hamiltonian gap	0.0012786965
Integrated gradient upper bound	0.0034406857
Absolute identity discrepancy	1.3×10^{-9}

For one active side, $-R''(\delta) = \Lambda_0 e^{-\kappa\delta} \kappa \{2 - \kappa(\delta - p)\}$ by Proposition 4.1. On $[0.15, 1]$ it is positive exactly when $p > 1 - 2/\kappa$, and then both factors decrease in δ , so the curvature modulus is attained at $\delta = 1$. With $\kappa = 1.2$ the condition is $p > -2/3$. The script evaluates it for both sides at every recorded time node and admissible inventory-side pair; the smallest recorded margin $p + 2/\kappa - 1$ is 0.2850603872. At a state where the condition failed, Theorem 6.2 would not apply and the local gap would instead be bounded by Proposition 6.5.

The recorded minimum is a numerical node check; it does not alone enclose the solution between nodes. The complete-interval residual enclosure in the next subsection is a separate certified calculation. The check uses this endpoint formula at each evaluated time and state. Numerical quadrature of the bound is not a formal interval-arithmetic certificate; the purpose is to test the performance identity and its implementation independently of the backward policy evaluation. No exchange data enter this exercise.

12.2 An exact enclosure for a computed HJB approximation

A separate two-state quadratic control specialization tests Theorem 7.1. Set $T = 1$, $A = [0, 1]$, $\Phi = 0$, and

$$r_x(a) = \beta_x a - \frac{1}{2}a^2, \quad k_{x,1-x}(a) = \lambda_x + d_x a, \quad x \in \{0, 1\}.$$

Here a is a control coordinate for the algebraic test; no calibration of an offset-demand curve is asserted. Use

Parameter pair	True	Estimated
(β_0, β_1)	(0.40, 0.60)	(0.42, 0.58)
(λ_0, λ_1)	(0.30, 0.40)	(0.31, 0.41)
(d_0, d_1)	(0.20, 0.10)	(0.22, 0.12)

The supplied approximation and implemented policy are

$$v(t) = (0.09(1-t) + 0.005, 0.17(1-t) - 0.005), \quad \tilde{a}_x(t) = \hat{\beta}_x + \hat{d}_x(v_{1-x}(t) - v_x(t)) + 0.01.$$

Both implemented actions stay strictly inside $[0, 1]$. The extra 0.01 produces a known nonzero first-order solver residual $\chi_x = 0.01$.

Writing $u = 1 - t$, each computed HJB residual is a quadratic polynomial in u . Its maximum absolute value on $[0, 1]$ is obtained from endpoints and any interior critical point. Exact rational arithmetic gives the uniform bound $\rho = 441709/12500000 = 0.03533672$. The terminal error is $d_T = 0.005$; $\text{span}(v) \leq 0.01 + 0.08u$; the primitive budgets are

$$\epsilon_{r,0} = \epsilon_{r,1} = 0.02, \quad \epsilon_{k,0} = 0.03, \quad \epsilon_{k,1} = 0.02, \quad \epsilon_{r,2} = \epsilon_{k,2} = \bar{K}_2 = 0,$$

and $\bar{K}_1 = 0.24$ is valid. Both Hamiltonians have exact curvature one for every continuation vector. Formula (32) therefore has the enclosure

$$D(t) \leq 0.005 + (0.0203 + \rho)u + 0.0012u^2.$$

Using this polynomial upper bound in b_1 and integrating (33) by rational arithmetic gives:

Quantity	Model units
Computed model-and-solver upper bound	0.001136664825
Direct true-model policy regret	0.000365316014
Forward-backward policy-value discrepancy	9.0×10^{-15}

The upper bound is constructed before using the true optimal value. Numerical true-model HJB and independent forward policy evaluation are used only for the displayed comparison. Unlike a sampled Hessian test, the residual supremum and bound integral in this quadratic example are enclosed over the complete time interval.

The supplement also checks the sharp margin exponents and the nonidentification construction. With $\lambda_0 = 1$, the latter has the analytical lower bound 0.00114336; direct scalar minimax optimization gives approximately 0.00828680. These are losses in the stated static criterion, not annualized trading returns.

12.3 Discounting, two-quote learning, and unbounded rewards

Stationary two-state policy iteration is checked at discount rates 0.2, 1, 3 against direct linear policy evaluation and the discounted occupation integral computed with matrix exponentials. The largest performance-identity discrepancy is below 3×10^{-16} . Computed-value and quote-optimization perturbations also satisfy Corollary 7.2 in these examples.

For the two-quote experiment above, numerical Poisson probability sums at $\tau = 20,000$ give τ times expected loss approximately 3.196429 and 1.597832 for $\kappa = 2, 3$, respectively. Their asymptotic limits are 3.187709 and 1.595807. The probability sums use explicit tail cutoffs; no simulated sampling error is involved, although their arithmetic is floating point. The exponential-mark moment formula is separately checked by quadrature. A noisy binary signal with a deterministic spread-reward component has strictly positive decision value and satisfies the unbounded-reward information ceiling. These checks exercise the corrected units and the post-calibration loss criterion.

12.4 Queue mechanics and event-law checks

The queue product in Proposition 3.4 is compared with a matrix resolvent in eight configurations, and the finite-horizon absorption probabilities are compared with 24 Erlang calculations. A six-state replacement model has two quote levels, two ranks per level, and absorbing fill and invalidation states. Market-order rates are 1.4, 0.7, net fill rewards are 0.45, 0.8, and cancellation of the one lot ahead has rate 0.2. Invalidation has rate 0.25. The decision clock has rate 2.5; replacement joins the other queue at rank one and costs 0.02. Active orders have running cost 0.02, horizon $T = 2$, and zero terminal reward.

Started at the first level with rank zero, the optimal value is approximately 0.35606137. Always replacing at each opportunity loses approximately 0.17792893. An independent forward occupation integral reproduces this loss within 5×10^{-11} ; forward evaluation of the maximizing feedback agrees with the HJB value within 2×10^{-14} .

The event reward and its first two derivative-error budgets are separately enclosed using rational polynomial coefficients on the whole action interval $[0, 1]$. These are exact coefficient bounds, not pointwise grid tests. Six pure-birth truncation examples compare Poisson exit tails with first-exit-time integrals. They illustrate the boundary allowance in Proposition 3.13; they do not estimate boundary probabilities for a fitted exchange model.

12.5 A continuing queue-and-inventory certificate

The next calculation applies Theorem 7.3 to a continuing policy with discrete messages. The state is $x = (z, q, s, \ell, n)$. The observed flow regime is $z \in \{0, 1\}$, inventory is $q \in \{-2, -1, 0, 1, 2\}$, and at most one unit order is active. Its side is $s = +1$ for a bid and $s = -1$ for an ask, its level is $\ell \in \{0, 1\}$, and its rank is $n \in \{0, 1\}$. An active order is allowed only when $-2 \leq q + s \leq 2$; each (z, q) also has one inactive state. These restrictions give 74 states.

This cell has an explicit regenerative queue convention. Each side and level contains one external lot. Submission or replacement puts the tagged unit behind that lot. When an ahead lot departs, an external lot is replenished behind the tagged unit; the tagged rank becomes zero. Thus replacement of a front order loses priority even at the same level. Replenishment is part of the declared compound event, not an inference from displayed depth. This is an idealized finite queue model.

Regime changes occur at rates 0.35 from zero to one and 0.55 in the reverse direction. They retain inventory and the tagged order. The base market-order rates and net fill rewards are

Regime z	Level ℓ	Base rate	Net fill reward
0	0	1.2	0.16
0	1	0.6	0.29
1	0	2.2	0.03
1	1	1.0	0.20

Multiply the base rate by 1.04 on the ask side and 0.96 on the bid side. At rank one, ahead departure has rate $\mu + 0.25$; at rank zero, filling has rate μ , changes inventory to $q + s$, and makes the order inactive. Invalidation at rate 0.20 makes an active order inactive without changing inventory. Only fills pay event rewards. Their rewards already include offset capture and simultaneous price marks; an offset and mark decomposition can be chosen with a compensated reference price and these net rewards.

The running inventory cost is $0.03q^2$, discounting is $\beta = 0.6$, and message opportunities have rate $\nu = 3$. Messages are keep, cancel at fee 0.002, submit an admissible side and level when inactive at fee 0.001, or replace an active order at fee 0.004. Submission and replacement reset rank to one. All messages preserve the inventory reservation.

Every external event rate lies within 0.0002 of its displayed center; every nonzero fill reward lies within 0.00001 of its center. Other rewards, the running cost, and the message law are known. The rectangular uncertainty set allows independent event perturbations in each state. Throughout it, $\Lambda = 6.5$ bounds external exits plus the message rate, and absolute reward rates are below one.

First compute a policy for the center model. Construct a policy that omits tagged rank by using, at every rank-one state, the message selected at the corresponding rank-zero state with the same z, q, s, ℓ . Message names are transferred, with their actual destination maps and fees. Elsewhere retain the center policy. The resulting decisions differ in 12 states. Starting inactive with $z = 0, q = 0$, the calculations give

Quantity	Value
Center-model optimal value	0.07701052
Center-model loss from omitting rank	0.00016417
Uniform gap bound $\ G\ _\infty/\beta$	0.06911853
State-dependent loss upper bound	0.000169512560
States with certified zero message gap	62

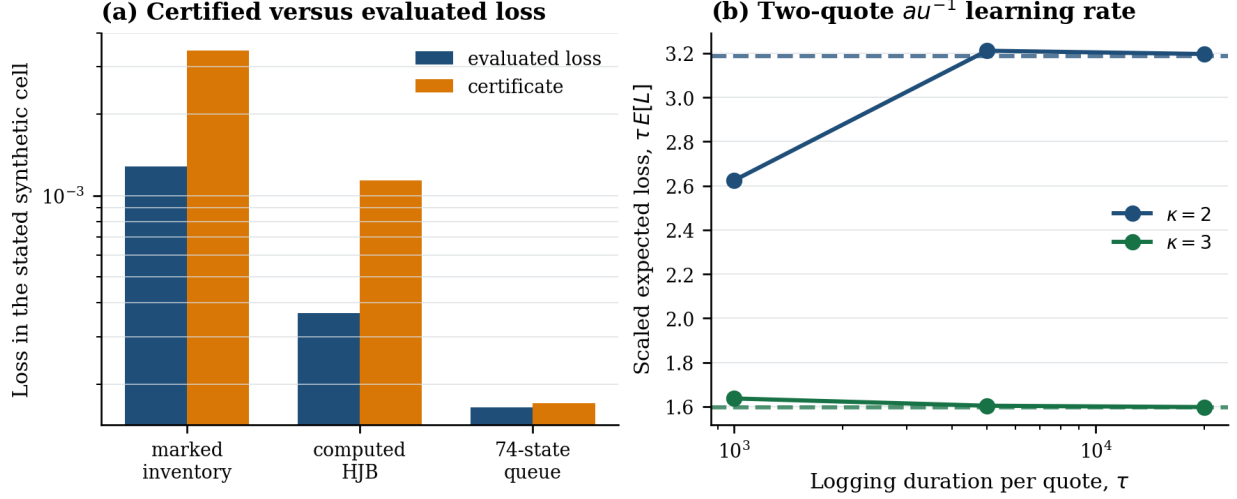


Figure 4: Certificates and learning rates in separate synthetic cells. Panel (a) compares independently evaluated policy loss with its applicable certificate for the marked-inventory model, the computed HJB model, and the 74-state queue model; the logarithmic scale is shared, but each pair retains its own declared model units. Panel (b) shows exact Poisson-sum evaluations of $\tau \mathbb{E}[L]$ for $\kappa = 2, 3$, converging to the constants in (45) (dashed lines). No simulated sampling or market data are used.

The last loss bound holds for every model in the stated uncertainty box. It is obtained using 320 Bellman enclosure updates and 320 fixed-policy occupation updates, all with exact rational arithmetic and outward rounding to 12 decimal places. The largest final value interval width is below 0.000166. Independent floating-point policy iteration and linear evaluation check the center and two interior perturbed models; their actual losses are respectively 0.0001641655, 0.0001641392, and 0.0001641918. The rational comparison supplies the enclosure; these three numerical evaluations do not stand in for a proof over the box.

The same supplement checks the exact rank-information lower bound and the Poisson replacement boundary. In the latter example $n = 2, R = 1/2, \vartheta = 1, \mu_0 = 1$, with parameter interval $[0.8, 1.2]$. For $z = 0.75$ in (47), the scaled losses at calibration durations $10^2, 10^3, 10^4, 10^5$ are 0.01560, 0.01583, 0.01590, 0.01592, approaching the analytic limit 0.01593474. These are synthetic decision experiments; no exchange data enter their construction.

Figure 4 places the three independently evaluated policy-loss checks beside the finite-sample approach to the two-quote learning limit.

12.6 Partial fills during a pending cancel

Use the compound-consumption example of Corollary 3.8, initially with three lots ahead and a two-lot bid. Keep states (n, m) with $n \in \{0, 1, 2, 3\}$, $m \in \{1, 2\}$, and one fully filled state $(0, 0)$. Inventory is $q = 2 - m$, so the live reservation satisfies $q + m = 2$. These nine states describe the order during one pending cancellation; no new orders are submitted.

Aggressive arrivals have rate 1.4, with batch sizes one and three of probabilities 0.7 and 0.3. The bid price remains fixed at 99.98. The marking reference is $S(2) = 100$, $S(1) = 99.99$, and $S(0) = 99.84$, where its argument is the remaining tagged quantity. These are declared state valuations in this finite episode. The offset before each event is therefore state dependent: $\delta_m = S(m) - 99.98$. Every

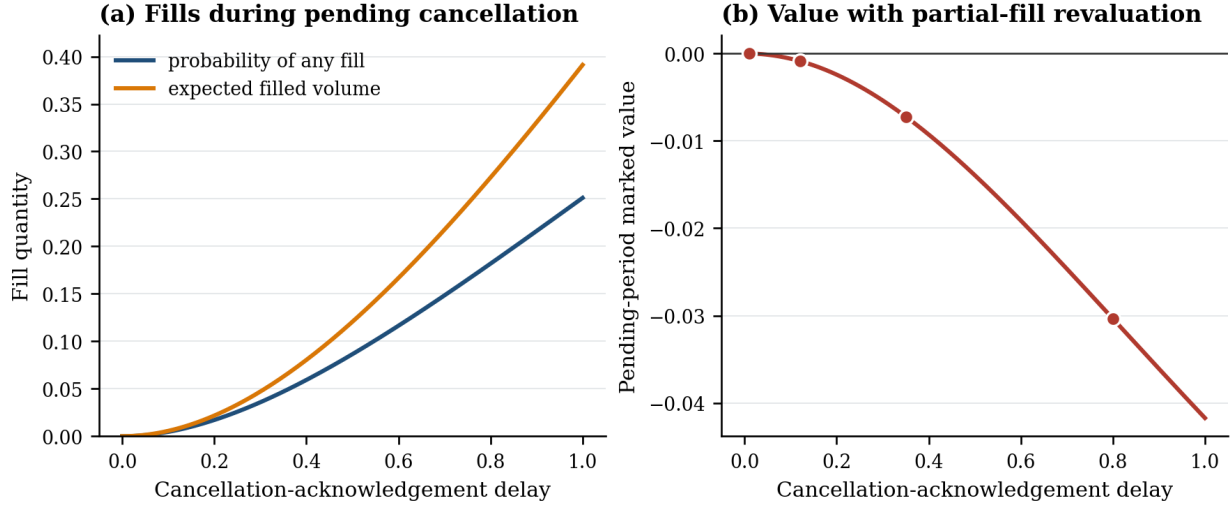


Figure 5: A still-live order during cancellation latency. Starting with three lots ahead and two tagged lots, panel (a) evaluates the compound Poisson law for the probability of any fill and expected filled volume. Panel (b) evaluates the exact pending-period value V^d , including the revaluation of earlier partial fills and post-acknowledgement inventory. The markers are the four delays reported in the table. All quantities come from the declared nine-state synthetic queue cell.

consuming event uses its full wealth increment

$$g_e = f\delta_m + (q + f)\{S(m - f) - S(m)\}.$$

This includes revaluation of shares filled earlier. A first one-lot fill increases marked wealth by 0.01; a subsequent fill of the remaining lot changes it by -0.29 . A two-lot sweep directly from the initial inventory changes it by -0.28 . The same final cash and marked inventory are obtained, while discounting and running costs can distinguish the timing. The generator thus retains the association between consumed volume, executed quantity, and the complete price mark. The running inventory cost is $0.03q^2$, the discount rate is $\beta = 0.8$, and the post-acknowledgement value is the specified inventory penalty $w = -0.05q^2$. While acknowledgement is pending, that running cost continues even if the order has already filled completely.

Delay d	Probability of any fill	Expected filled lots	Pending value V^d
0.01	0.00004958	0.00005858	-0.00000591
0.12	0.00655206	0.00808693	-0.00087478
0.35	0.04702895	0.06281535	-0.00722870
0.80	0.18180674	0.27252734	-0.03035474

The probability of any fill and the expected filled volume are different risk quantities. The table's marked value also includes the dependence of fill quality on batch size and the inventory remaining after cancellation. Instantaneous cancellation would give zero value at the initial state.

Figure 5 traces these quantities over the complete delay interval rather than only at the four tabulated delays.

At $d = 0.35$, the residual constants of Theorem 3.11 are exactly

$$G = \frac{303}{500}, \quad B = \frac{3473}{2500}, \quad C = \frac{38763}{12500}.$$

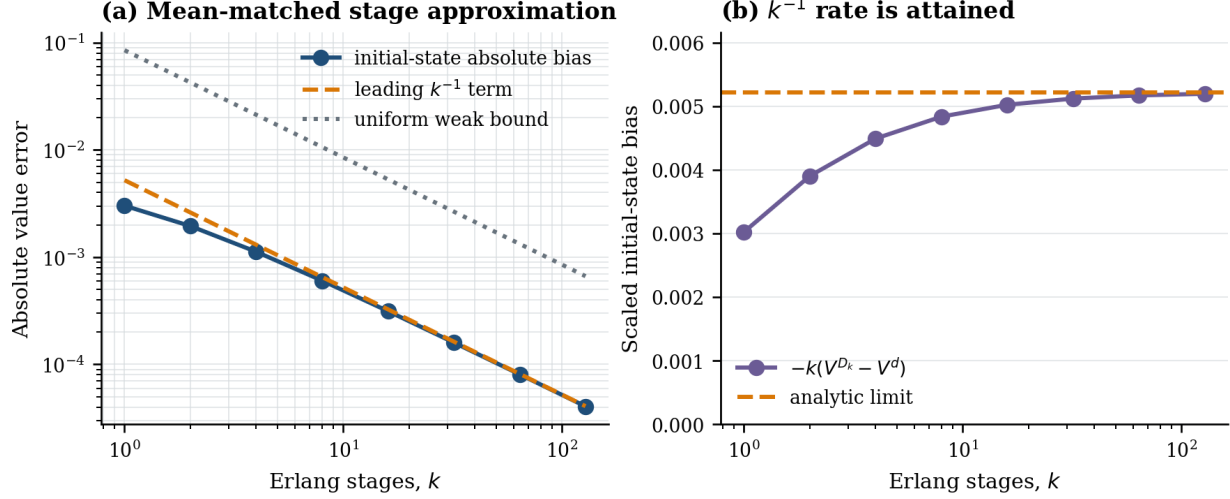


Figure 6: Mean-matched Erlang approximation of deterministic cancellation latency at $d = 0.35$. Panel (a) compares the initial-state absolute bias with the leading k^{-1} term and the uniform weak bound from Theorem 3.11. Panel (b) shows that $-k(V^{D_k} - V^d)$ approaches the analytic coefficient 0.00521944. The independent stage recursion and matrix calculation use the same nine-state synthetic cell as Figure 5.

The uniform expected-value approximation bound is $Bd^2/(2k) = 0.08508850/k$.

Erlang stages k	Initial-state bias $V^{D_k} - V^d$	Uniform absolute bound
4	-0.00112257	0.02127213
16	-0.00031387	0.00531804
64	-0.00008077	0.00132951
128	-0.00004058	0.00066476

The scaled initial-state bias tends to -0.00521944 , the coefficient $d^2(e^{dA}Ag)_x/2$ in (19). The uniform bound covers all nine initial states and need not be attained at the displayed one.

Figure 6 shows both the finite-stage error and the sharp asymptotic normalization.

The supplement compares the partial-fill distributions with direct compound-Poisson sums in 32 cases and their generating functions at 128 parameter points. Their largest distribution discrepancy is below 2×10^{-15} . Matrix-exponential cancellation values agree with the residual integral to below 2×10^{-16} . Erlang stage recursions are independently compared with matrix resolvents and integration over the gamma density. The constants G, B, C use exact rational arithmetic; the reported matrix values are floating-point evaluations of the declared synthetic model.

13 Beyond a distance-only fill law

A dealer HJB is an optimization equation, not a unique specification of market behavior. The omitted mechanisms belong to a particular state, event law and objective. The original Avellaneda–Stoikov model uses exponential utility, not merely a quadratic inventory penalty (Avellaneda and Stoikov, 2008). A formula involving $1/k$ comes from a particular exponential intensity cell; it is not a universal competitive spread. For example, maximizing $Ae^{-k\delta}(\delta - c)$ gives the unconstrained

interior condition $\delta = c + 1/k$ only when the continuation-adjusted charge c is held fixed and the quote is continuous. Tick constraints, endogenous marks, queue resets and competitors can change that calculation.

13.1 The six requirements for a continuing quote

The modelling requirements of [Noguer Alonso \(2026, Section 37.1\)](#) take a specific form here. *Fill law*: tagged rank, partial fills, reservations and pending cancellation are explicit in Sections 3.3, 3.5 and 3.7; venue priority rules and the joint law of feed and order latency are not. *Information*: the marked-wealth charge (50) must use the joint law of fill and price mark, and Proposition 13.1 shows exactly what is lost when only its marginals are known. *Objective*: the certificates concern the additive marked-wealth objective; drawdown, VaR or Sharpe criteria need a wealth-and-peak state or a separate formulation, and an adaptive mean–variance criterion needs a precommitment or equilibrium interpretation. Impact and liquidity, competition and evaluation carry over as stated there; a reduced-form response to rival depth is not a Nash equilibrium, and a replay that holds the historical book fixed cannot validate the book’s response to a different quoting policy.

13.2 One event, one marked-wealth charge

Let a joint event mark contain executed quantity ξ (positive for a buy), execution price P , the reference-price jump J , cash fee f (negative for a rebate), and any queue or message update. With pre-event inventory q and reference S , the exact wealth increment is

$$\Delta W = \xi(S - P) + (q + \xi)J - f. \quad (50)$$

Indeed cash changes by $-\xi P - f$, and marked inventory changes from qS to $(q + \xi)(S + J)$. Thus a finite-activity event kernel $\nu_t(dz \mid x, a)$, a between-event reward rate r_0 , and a between-event generator $\mathcal{L}_0$ give the schematic HJB

$$0 = \partial_t v + \sup_{a \in \mathcal{A}_t(x)} \left\{ r_0 + \mathcal{L}_0 v + \int [\Delta W(x, a, z) + v(t, F(x, a, z)) - v(t, x)] \nu_t(dz \mid x, a) \right\}. \quad (51)$$

Here v values *future increments*, excluding already booked wealth; running penalties belong in r_0 . This convention prevents subtracting a fill markout twice. If a reward instead books a price mark at a future horizon, the overlapping continuation must be adjusted, or the state must retain the outstanding markout obligation. The shared Bellman step requires a consistent accounting boundary.

13.3 Sharp ambiguity from missing fill–jump dependence

Consider one ask quote at spread d , initial inventory zero, no continuation or terminal inventory penalty, and a fee f per fill. Let $X \in \{0, 1\}$ indicate its fill and $J = hY$, where $Y \in \{0, 1\}$, $h > 0$, is a simultaneous upward price jump. A nonparticipating dealer earns zero in this cell. Its quoting gain is

$$G = \mathbb{E}[X(d - f - hY)].$$

The endpoint liquidation convention is marked wealth, not cost-free future liquidation. It is deliberately the same for both actions.

Proposition 13.1 (A sharp marginal-only quoting interval). *If $\Pr(X = 1) = p$ and $\Pr(Y = 1) = q$, but their dependence is unknown, then the exact set of possible quoting gains is*

$$[p(d - f) - h \min(p, q), \quad p(d - f) - h \max(0, p + q - 1)]. \quad (52)$$

Independence selects the single value $p(d - f) - hpq$. A nonnegative lower endpoint is necessary and sufficient for this quote to weakly improve on nonparticipation for every joint law with these marginals.

Proof. Write $z = \Pr(X = 1, Y = 1)$. The four cell probabilities are $z, p - z, q - z, 1 - p - q + z$. Their nonnegativity is equivalent to $\max(0, p + q - 1) \leq z \leq \min(p, q)$. Every point in this interval is a joint law, and gain is the decreasing affine function $p(d - f) - hz$. Its endpoint values give the interval and the comparison with zero. $\square$

The argument is the two-point marginal-coupling calculation of [Noguer Alonso \(2026, Proposition 37.1\)](#); the proposition records its market-making instance, in which the participation decision rather than a routing decision is reversed.

Take $p = 0.40, q = 0.25, d = 1, f = 0, h = 2$, in tick units. Independent marginals suggest a gain of 0.20 ticks per opportunity. The admissible joint law with all upward jumps occurring at fills has gain -0.10 . Fill probability and unconditional jump probability are identical. Their full ambiguity interval is $[-0.10, 0.40]$. With probability $0.10 + 0.15\theta$ assigned to the joint fill/jump event, gain is $0.20 - 0.30\theta$: the quote becomes unprofitable once $\theta > 2/3$.

Table 2: Dependence changes participation despite fixed marginal fill and jump probabilities. Values are synthetic ticks per opportunity.

Law	Quoting gain	Selected action
Independent	0.20	Quote
Adverse coupling	-0.10	Inactive
Marginal-only worst case	-0.10	Inactive

For partial fills and several marks, a finite coupling matrix gives the same kind of exact stage calculation. Given marginal weights p_i, q_j and a continuation-aware event gain g_{ij} , its worst expectation is

$$\min_{\pi \geq 0} \sum_{ij} \pi_{ij} g_{ij}, \quad \sum_j \pi_{ij} = p_i, \quad \sum_i \pi_{ij} = q_j. \quad (53)$$

The event destinations and continuation enclosure must be included in g_{ij} ; optimizing only instantaneous spread is insufficient. This is a finite marginal-coupling linear program, not a new solution of the full dealer game. To feed a stage bound into [Section 8](#), retain the same underlying model in candidate and baseline advantages. Independent minimization of their values can discard cancellation of common uncertainty.

The script `verification/realism_stress.py` checks the interval against independent linear programs over a probability grid, including degenerate marginals, and verifies the mixture switch. These exact toy laws demonstrate an identification obstruction; they are not inferred from market observations.

13.4 What would make the added realism measurable?

The evaluation unit should be an order/message episode with observable queue history, submitted size, executions, cancel acknowledgements and price marks at preregistered horizons. Preserve

causality using receipt times available to the agent. Split by time and keep overlapping marks within the same partition. Estimate partial-fill distributions and the joint fill/mark law conditional on rank, size, latency and regime, with uncertainty increasing where action support is sparse. Compare a frozen baseline with the same net fee and terminal inventory accounting. Report net marked reward, inventory and drawdown tails, stale-quote exposure, fill calibration and certificate coverage. An ablation that adds each variable separately should be accompanied by combined latency, toxicity and competitor-response stress. Neither a positive simulated spread nor a narrow in-model interval supplies empirical profitability.

14 From a quoting certificate to an executable message policy

An academic benchmark and a production policy differ through their state, action, observation and implementation contracts. Continuous time is not itself the obstruction: a continuous-time jump process can retain integer shares and tick prices. Nor does a Hawkes model impose independent arrivals; its purpose is to represent excitation and temporal dependence (Bacry et al., 2015). This section makes the additional contracts explicit and derives a finite-action implementation certificate.

14.1 A joint law for messages, fills and price marks

At a local decision time t , let $\mathcal{G}_t$ contain only fully available market data, reconciled private reports and prior sent messages. A sufficient state can include tick spread, tagged ranks, inventory, outstanding buy and sell reservations, pending requests, flow features, signal age and a posterior for hidden regime or liquidity. The control is a native message, selected from a finite feasible set: retain, cancel, replace, submit at an allowed tick and size, or take liquidity. The action mask also enforces price, inventory and message-budget constraints.

The required conditional law is jointly over fill times, executed quantities, execution prices, fees, subsequent price marks, effective cancellation, private-report arrival and next observation. Cancellation effectiveness and its report time are distinct. A local cancel request stops neither fills nor inventory risk by definition. State-dependent congestion can couple a delayed cancel to precisely those events that make its order unattractive. Such a law is covered by an augmented-state kernel, not by multiplying independent marginal latency and toxicity distributions.

Maintain $q + o^b \leq Q_{\max}$ and $q - o^a \geq -Q_{\max}$, where o^b, o^a include unresolved submissions and requests that can leave orders executable. A buy fill of f changes (q, o^b) to $(q + f, o^b - f)$, preserving the first sum and relaxing the second bound; a sell fill has the symmetric property. A cancel request does not release the reservation. This gives a pathwise inventory invariant provided reports are reconciled and terminal release is authoritative. A risk layer that merely refuses to count a real fill at a boundary is not an inventory constraint.

Nasdaq’s public ITCH book and private OUCH lifecycle have different roles (Nasdaq, 2026a,b). An order-by-order feed does not expose hidden resting orders or the dealer’s complete private state. A simulated FIFO rank is exact only within the declared visible queue and arrival convention. NYSE’s parity allocation requires a different transition map (New York Stock Exchange, 2026). Exchange-specific matching, repricing and replacement rules must therefore be versioned inputs to the model.

14.2 Ticks and implementation certificates without smoothness

Fix a state and decision horizon. Let $Q(a)$ be the true action value with one common continuation policy or optimal continuation, and let $s(a)$ be the score actually computed by the implementation

for each feasible message. Suppose $|Q(a) - s(a)| \leq e(a)$. The bounds may differ by message.

Lemma 14.1 (Certified message selection). *For any emitted feasible action $\tilde{a}$,*

$$\max_a Q(a) - Q(\tilde{a}) \leq \max_a \{s(a) + e(a)\} - \{s(\tilde{a}) - e(\tilde{a})\}. \quad (54)$$

For a baseline action a_0 , its one-step advantage satisfies

$$Q(\tilde{a}) - Q(a_0) \geq s(\tilde{a}) - s(a_0) - e(\tilde{a}) - e(a_0). \quad (55)$$

If all errors are at most e and the emitted action is within ξ of maximizing s , its loss is at most $2e + \xi$. If one score exceeds all others by more than $2e$, its exact maximizing action is identified.

Proof. Bound each competing value above by its score plus its error, and the emitted value below by its score minus its error. Subtraction proves the first claim. Reverse the two bounds for the baseline comparison. The uniform bound and strict-margin conclusion follow immediately. $\square$

This result applies to integer ticks and discrete order instructions with no differentiability assumption. For a dynamic guarantee, choose the common continuation as the fitted optimal value and insert the resulting local loss into [Noguer Alonso \(2026, Theorem 39.1\)](#), or use the earlier baseline-policy performance identity with that baseline's continuation. A myopic score against a zero continuation cannot certify improvement of an infinite-horizon inventory policy.

A deployable score enclosure may combine verified interpolation error, fixed-point arithmetic error, model/continuation error and feature staleness. For example, if the score is L_a -Lipschitz in a numerical feature z , its stored representative $\bar{z}$ satisfies $\|z - \bar{z}\| \leq h$, and arithmetic error is at most ν_a , these components contribute $L_a h + \nu_a$. This statement does not apply across unmodeled changes of queue rank or admissibility. Such branches must be enumerated or enclosed separately. Integer arithmetic also needs a proved range that excludes overflow; a mathematical quantization tolerance alone does not establish it.

If an omitted event set has conditional probability at most p_{tail} and action payoff on it has absolute error at most D , add Dp_{tail} to the expected-score allowance. For unbounded losses one instead needs an integrable tail bound, for example $\mathbb{E}[|\text{error}|1_{\text{tail}}]$. Quoting a measured latency percentile without either allowance does not bound expected policy loss.

14.3 The economic value of latency tails

Proposition 14.2 (Mean latency does not determine expected loss). *A resting order incurs loss $L > 0$ if it is still live when an adverse trade arrives at a fixed time $\tau > 0$, and has no other payoff. A cancellation effective exactly at τ is processed before that trade. Its cancellation becomes effective at an independent delay $D \geq 0$ with $\mathbb{E}D = m$, where $0 < m < \tau$. Then the expected loss is $L \Pr(D > \tau)$, and over all such laws*

$$\{L \Pr(D > \tau) : D \geq 0, \mathbb{E}D = m\} = [0, Lm/\tau).$$

The value 0 is attained by the constant delay $D \equiv m$; the supremum is approached by two-point laws on $\{0, \tau + \epsilon\}$ and is not attained.

Proof. The order is live at τ exactly when $D > \tau$. If $\Pr(D > \tau) > 0$ then $m \geq \mathbb{E}[D1_{\{D > \tau\}}] > \tau \Pr(D > \tau)$, so $\Pr(D > \tau) < m/\tau$. The law putting mass $m/(\tau + \epsilon)$ at $\tau + \epsilon$ and the rest at zero has mean m and $\Pr(D > \tau) = m/(\tau + \epsilon) \rightarrow m/\tau$. The set of laws with mean m is convex and the probability is linear in the law, so the attainable set is an interval containing 0 and all values below m/τ . $\square$

In the cell with loss $L = 3$ ticks, adverse trade at $\tau = 2$ and mean delay $m = 0.5$, the constant delay gives 0, the mixture of 0.1 (probability 0.9) and 4.1 gives expected payoff -0.3 ticks (loss 0.3 ticks), and delays with the same mean can reach losses arbitrarily close to 0.75 ticks. Clock units are synthetic.

More generally, for an adverse trade at time τ with fixed loss $L > 0$ and no other payoff, the expected loss is $L\mathbb{P}(D > \tau)$ under the specified joint law. If τ is random and correlated with D , the probability is joint; replacing it by an independent-latency calculation changes the model. The earlier phase-type expansion remains useful under its assumptions, while this simple tail example identifies an effect that moment matching alone cannot control.

FPGA acceleration is one implementation option, not a theorem hypothesis. The relevant engineering record is the complete latency path: receipt, decode, feature update, risk check, policy evaluation, serialization, transport, exchange processing and private-report return. Tail behavior under bursts and degraded conditions matters alongside central tendency. A lookup can be constant in operation count while cache misses, contention or queueing make wall-clock delay variable. Expensive estimation and optimization may run outside the order-response path, provided the cached policy’s age and approximation errors remain in its declared envelope.

14.4 Calibration with competing events and policy selection

For a tagged order, log its entry context, size, rank estimate, message history, every partial fill, effective terminal event and subsequent markouts. A simplified cause-specific likelihood with observed at-risk indicator Y_t and event hazards $\lambda_j(t \mid \mathcal{G}_{t-}, a)$ is

$$\ell = \sum_j \int \log \lambda_j(t) dN_j(t) - \int Y_t \sum_j \lambda_j(t) dt.$$

This is a counting-process likelihood under its declared conditional hazards, not an instruction to discard unfilled orders. A cancellation is an observed competing lifecycle event for the deployed policy. Estimating the hypothetical fill time had the order not been canceled additionally requires conditional censoring assumptions and support for remaining at risk. Conditioning only on filled orders estimates a selected population and cannot identify an unconditional fill hazard.

Model assessment should check fill time, partial volume, price mark and cancellation exposure jointly across supported quote actions. Calibration near a replacement boundary is especially consequential because small value errors can change a discrete message. Use chronological holdouts and explicit regime or drift allowances; do not apply an independent-sample confidence formula to a clustered event stream without justification. The finite calibration construction in Section 15 states one sufficient sampling experiment and an explicit unsupported-row rule.

14.5 Reference implementation and falsification cases

The code emits a normalized event trace with exchange and receipt times, tracks a tagged FIFO queue through adds, cancels ahead/behind and partial consumption, and retains executable exposure until cancellation is effective and privately resolved. It exercises the following failure cases: treating a cancel request as terminal, releasing exposure before reconciliation, using a future exchange event before receipt, double-counting a report, changing quote price without accounting for lost rank, and choosing a compiled message whose score margin is smaller than its error allowance.

The independent policy evaluation tests do not assume the selected action is optimal. They compute its value under the test law and compare its loss with the certificate. Message conformance, real latency measurements, borrow/funding access, actual fee tiers and production incident recovery

Table 3: Synthetic quoting checks. Payoffs are ticks per declared order; clock units are arbitrary. The mean delay is identical in the two rows.

Quantity	Value
Mean cancellation delay, both laws	0.500000
Constant-delay payoff (ticks)	0.000000
Mixture-delay payoff (ticks)	-0.300000
Compiled-score comparisons	1,000
Random finite-model comparisons	200

remain external evidence requirements. The mathematical completion here is an explicit model-to-implementation contract with reproducible checks, not an assertion of a profitable exchange-connected market maker.

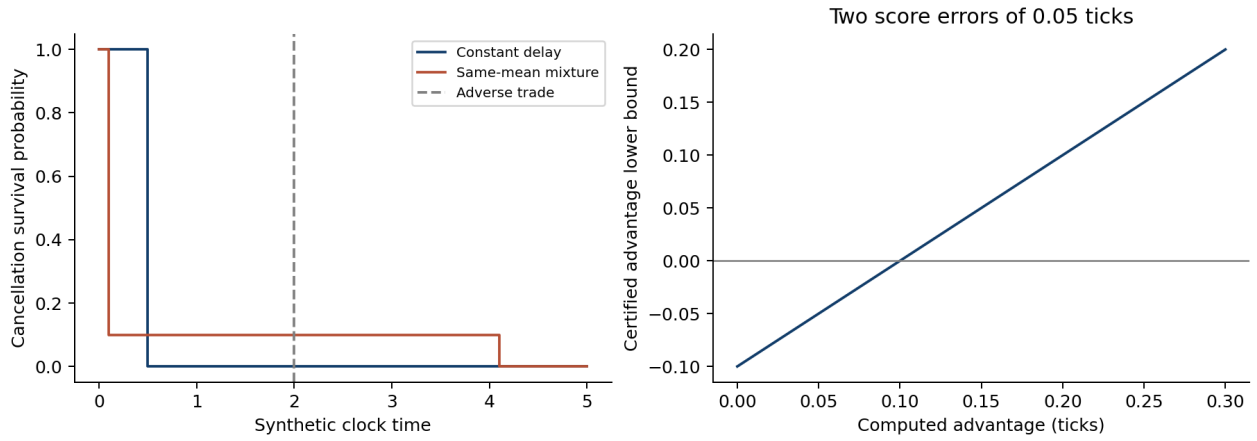


Figure 7: Discrete implementation diagnostics generated by the accompanying reference suite. All axes use synthetic model units; the curves illustrate the declared mechanisms rather than estimated market behavior.

14.6 Persistence, competition and the scope of the dealer law

The event rates of Assumption 3.1 depend on the observable state (q, z, a) , so persistent flow enters only through the regime coordinate z . The switching-friction mechanism of Rodriguez Dominguez (2026b) predicts execution-weighted duration restrictions that a stationary two-regime chain, as in Section 12.5, cannot reproduce asymptotically. A finite-window approximation must be justified and its error charged; enlarging a finite state alone does not create true asymptotic long memory. The equilibrium conclusions of Rodriguez Dominguez (2026a) rest on timing restrictions that differ from ours, since the conditional fill-price marks here can represent informed flow. Their relevance is to the family Θ of Theorem 8.1, which is fixed: if rivals come to condition on the same representation, Θ must contain the law after that adoption, not before it.

15 Calibration, implementation error, and the evidence boundary

The deployment interface is the one stated and proved in Nogueir Alonso (2026, Section 39). With a common initial law, observation class, feasible masks and terminal reward error at most η_N , suppose

the fitted rewards and next-state kernels satisfy $|r_k - \hat{r}_k| \leq \eta_k$ and $\text{TV}(P_k^a, \hat{P}_k^a) \leq \epsilon_k$ uniformly over feasible states and messages, and S_k bounds the span of fitted continuation values. Then, with $B_N = \eta_N$ and $B_k = B_{k+1} + \eta_k + \epsilon_k S_{k+1}$ every common feasible policy has $|J - \hat{J}| \leq B_0$, and a compiled policy with fitted action gaps ζ_k loses at most $2B_0 + \hat{\mathbb{E}} \sum_k \zeta_k$ (Noguer Alonso, 2026, Theorem 39.1). For prespecified state–action rows with independent draws, simultaneous radii follow from Noguer Alonso (2026, Proposition 39.2).

Here the reward includes the marked fill charge (50), and the kernel includes tagged-rank transitions and pending-cancel states. Theorems 7.3 and 8.1 are the discounted analogues and use value enclosures over a parameter box Θ rather than per-row radii. The row radii can populate Θ only if the rows are the state–message pairs of the finite queue model and the independent sampling assumption holds; a single dependent message stream does not satisfy it without a martingale, mixing or block argument. Replay and off-policy evaluation need the support, invariance and unconfoundedness conditions of Noguer Alonso (2026, Section 39.4); Proposition 10.1 is an instance of what fails without action coverage. Message conformance, measured latency, fee tiers and incident recovery remain external evidence requirements.

16 Scope and consequences

The accounting identity applies to general admissible semimartingales. The dynamic certificate uses a finite-state, translation-invariant model and an additive marked-wealth objective. Its quadratic form requires a common region with genuine curvature, whereas switching and tick choices are covered by a uniform-error bound and, when supplied, an occupation-margin refinement. The computable certificate also accounts for HJB, terminal, and optimization errors. The shutdown threshold concerns incremental gross markouts and does not override inventory constraints or continuation values.

The event construction identifies the required queue and message state, retains conditional fill marks, and maps event-law errors into the control budgets. Replacement has a continuation-value cost because it changes rank. A capped model also needs the separate exit allowance in Proposition 3.13. These modeling terms remain present even if its numerical HJB is solved exactly. The interval message certificate accommodates discrete queue decisions directly and retains their state occupation. Its numerical example encloses an entire declared rate-and-reward box, with explicit outward rounding. A model family that changes message maps or the observation class requires an additional comparison. Batch fills preserve the reservation pathwise but change both the execution-volume law and the conditional mark contribution. The cancellation-delay identity retains these fills until acknowledgement. Its finite-stage approximation has an explicit, sharp expected-value error bound under the stated independent-delay law; dependence between congestion and future flow requires a joint state model.

Three objects must be distinguished in calibration: the rate of fills, the conditional law of their marks and destination states, and the value function induced by those primitives. The certificate makes all three channels visible, including the effect of an estimated value function. The numerical supplement checks algebra and a synthetic finite-state control problem, including a globally enclosed rational example. Action coverage is a separate identification requirement: the fixed-quote counterexample prevents vanishing statistical guarantees from being inferred from sample size alone. The rank lower bound shows a distinct information obstruction even when every primitive is known. With rank observed, the replacement-boundary experiment has a matching minimax rate of order $\tau^{-1/2}$, whereas smooth strongly curved choices can have quadratic estimation loss. The two-quote construction supplies a positive learning guarantee for a specified exponential cell, including an

unknown intensity scale. Discounting removes the terminal horizon while retaining dependence on an effective horizon and the curvature modulus. The signal-value extension replaces bounded losses by a uniform centered moment condition. The production extension adds compiled-score error bounds and applies the monograph’s simultaneous finite-sample construction under explicit independent row-sampling assumptions. These bounds account separately for model error and implemented action loss. They do not identify a market fill law from unsupported actions or validate independent sampling for dependent exchange observations.

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